AI-Assisted Dental Clinic Management System

Developing an AI-Assisted Dental Clinic Management System with Predictive Appointment Scheduling and Patient Record

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Chapter 1: Introduction

1.1 Project Context

The delivery of quality dental care depends not only on clinical expertise but also on the efficiency of administrative operations. For many small and independent dental clinics in the Philippines, day-to-day management remains heavily reliant on manual, paper-based processes that have become increasingly unsustainable as patient volumes grow.

This project was developed in partnership with the client, a licensed dentist who owns and operates two private dental clinics. For this project, the system will be developed and implemented for one clinic only, as both branches operate independently with separate patient records and administrative workflows.

The clinic is managed day-to-day by a dedicated secretary whose primary responsibilities include scheduling patient appointments, reminding patients of their upcoming visits, and maintaining all physical patient records. When a new patient visits the clinic, the secretary first collects and records their personal information, dental history, and medical history on a paper

form. The patient is then directed to the operating area, where the dentist conducts an interview and proceeds with the examination and treatment.

Appointments are currently scheduled by phone or walk-in. Patients contact the clinic, the secretary manually checks a paper logbook for available slots, and the appointment is written by hand. The client enforces a strict punctuality policy; all patients are required to be on time, and those who anticipate being late are expected to send a message in advance. Despite this policy, the clinic still experiences scheduling issues. As the client confirmed during the interview, scheduling overlaps still occur even under the current system, with double bookings happening on occasion.

A particularly significant operational gap arises when a patient cancels or fails to show up for their scheduled appointment. As the client directly stated during the interview, when a patient does not attend, the slot simply goes to waste. There is no system in place to reassign the slot, notify a waiting patient, or take any corrective action. This results in lost consultation time and reduced clinic revenue.

Beyond scheduling, the clinic also faces a serious records management challenge. The secretary is responsible for maintaining thousands of physical patient folders stored in filing cabinets. Retrieving a specific record requires manually searching through these cabinets one by one, a tedious and time-consuming process. The volume of paper records has created a significant storage crisis, as the clinic continues to run out of physical space to accommodate new files. During consultations, the dentist and clinical staff must review these handwritten records while the patient is in the dental chair, record new findings by hand on physical dental charts, and update the patient's history in real time all on paper.

The client expressed strong interest in modernizing clinic operations, responding positively to the concept of an AI-assisted management system during the interview. The client specifically acknowledged the value of a predictive scheduling feature, noting that the ability to avoid overlaps and manage appointment slots more intelligently would be a significant improvement. The client also noted that going digital would make the secretary's workflow easier and shorter by removing the burden of manual follow-ups and physical filing.

The goal of this project is to develop an AI-Assisted **Dental Clinic Management System** that addresses these operational challenges by digitizing patient records and implementing a predictive appointment scheduling feature. The system aims to eliminate the storage crisis caused by paper records, reduce the rate of missed and wasted appointments through automated reminders and prediction, and provide the clinic with a smarter, more efficient way to manage its daily operations.

1.2 Statement of the Problem

This project seeks to address the following:

• **Inefficient management of cancelled and missed appointments:**

When patients cancel or fail to attend scheduled appointments, the clinic lacks an automated or structured process to manage the vacant time slot. As a result, cancelled appointments are not reassigned, leading to unused consultation hours and reduced clinic productivity.

- **Absence of a patient waiting list or notification mechanism:**

The clinic does not have a system to notify other patients when an appointment slot becomes available. Even when patients inform the clinic in advance of a cancellation, there is no mechanism to alert waiting patients, preventing optimal utilization of available time and delaying care for other patients.

- **Inefficient record management due to paper-based patient documentation:**

The clinic faces inefficiencies in managing patient records because personal information, dental history, and medical history are documented and stored exclusively on physical paper forms. This manual system limits organization slows down record handling, and increases the risk of lost, damaged, or outdated patient information.

- **Difficulty in generating reports:**

Due to the lack of a centralized digital record system, both the dentist and the secretary experience challenges in quickly retrieving comprehensive patient data. This can lead to inefficiencies during consultations, inconsistent documentation, and potential impacts on the quality and continuity of patient care.

1.3 Objectives

General Objective:

To design and develop an AI-assisted dental clinic management system that digitizes patient record management and optimizes appointment scheduling, with the goal of improving the operational efficiency of the clinic and the quality of care delivered to its patients within the nine-month project lifecycle.

Specific Objectives:

- To reduce unused consultation slots caused by cancellations and no-shows by at least 50% within the nine-month project lifecycle, as measured through the system's appointment tracking logs

The clinic currently has no mechanism to reassign cancelled slots, resulting in all cancelled appointments going to waste. By the end of the refinement phase, the system should reduce unused consultation slots caused by cancellations and no-shows by at least 50% compared to the baseline rate recorded during the first month of deployment, as measured through the system's appointment tracking logs.

- To enable clinic staff to retrieve any complete patient record within 3 minutes of initiating a search by the end of the refinement phase, as validated through user acceptance testing with the clinic secretary

The clinic secretary currently retrieves patient records by manually searching through physical filing cabinets, a process that has no defined time standard. By the end of the refinement phase, the system should enable clinic staff to retrieve any complete patient record within 3 minutes of initiating a search, compared to the manual process, as measured through user acceptance testing with the clinic secretary.

- To fully automate at least 3 manual administrative tasks (appointment reminders, record retrieval, and slot management) by the end of the refinement phase, as validated through secretary feedback during user acceptance testing

The secretary currently spends a significant portion of her working hours manually reminding patients, searching for records, and managing a paper logbook. By the end of the refinement phase, at least 3 of these manual tasks appointment reminders, record retrieval, and slot management should be fully automated by the system, as validated through secretary feedback during user acceptance testing.

- To successfully migrate and store 100% of newly registered patient records in a centralized digital database with automated cloud backup by the end of the development phase, as verified through system testing and client sign-off

The clinic currently stores all patient information on paper with no digital backup, making records vulnerable to loss and damage. By the end of the development phase, the system should successfully migrate and store 100% of newly registered patient records in a centralized digital database with automated cloud backup, as verified through system testing and client sign-off.

1.4 Significance of the Project

This project is significant to the following stakeholders:

To the Clinic Owner The system reduces time lost to administrative inefficiencies and provides better visibility over the clinic's daily schedule. Cancelled and missed appointment slots can be identified and reassigned quickly, reducing lost revenue and improving overall clinic productivity. The system also eliminates the risk of scheduling overlaps that currently occur under the manual logbook system.

To the Clinic Secretary As the primary user of the system, the secretary benefits most directly from this project. Tasks that previously required manual searching through filing cabinets,

handwriting appointment entries, and personally following up with patients can now be handled digitally making the administrative workflow significantly easier, faster, and more organized.

To the Patients Patients receive automated reminders before their appointments, reducing the likelihood of forgetting or missing their scheduled visits. They can also access their own profiles, view their appointment history, and receive notifications directly on their phones, improving their overall experience with the clinic.

To Future Developers and Researchers This project serves as a reference for future studies on AI-assisted clinic management systems, particularly in the context of small independent dental practices in the Philippines where digital adoption in healthcare administration remains limited.

This project also contributes to **SDG 3: Good Health and Well-Being** by improving the delivery of dental care through more efficient clinic operations, better patient engagement, and more reliable continuity of care.

1.5 Scope and Limitations

This project covers the design and development of a dental clinic management system for one clinic of x. The system includes the following:

- A digital patient record module for creating, retrieving, updating, and archiving patient information in a centralized and secure database.
- An AI (agentic) appointment scheduling module that predicts no-show risk based on historical appointment patterns, sends automated reminders to patients, and flags cancelled or at-risk slots for reassignment.
- Basic user authentication for the clinic secretary and authorized staff.
- A desktop application interface installable on the clinic computer.

The following are the limitation of this project:

- The system will be implemented for one clinic only and will not cover the second branch operated by the client, as both clinics maintain separate records and independent workflows.
- The system will not handle billing, invoicing, or payment processing.
- Integration with external health information systems or insurance providers is not included in this version.
- The predictive scheduling model's accuracy may be limited during early deployment due to insufficient historical appointment data.
- The system does not include treatment prediction or clinical decision support features, as these fall outside the administrative scope of the project.

Chapter 2: Review of Related Literature and Systems

2.1 Related Literature on Dental Clinic Management Systems

The digitization of dental clinic operations has been a growing area of research in recent years. Several studies have proposed and developed systems aimed at replacing manual, paper-based processes in small dental practices. Kok et al. developed a Dental Clinic Management System (DCMS) that allows patients to manage their own appointments, view their dental history, and receive feedback from dentists, demonstrating that transitioning from paper records to a digital platform significantly reduces administrative workload and improves overall clinic efficiency. Similarly, a study conducted for Ruth Aranez Dental Clinic presented a comprehensive clinic management system covering patient records, appointment handling, and decision support, validating that custom-built systems are a practical and effective solution for small independent practices. Building on this, another local study proposed a web-based dental management system with SMS notification and a decision support module for detecting tooth disease, further showing that integrating notification features into clinic systems improves patient engagement and reduces missed appointments.

These studies collectively establish that small dental clinics stand to benefit greatly from purpose-built digital management systems, and that the core features of patient record management and appointment scheduling are the most critical areas to address both of which are the primary focus of this project. The operational challenges observed in the clinic of , including physical paper records, manual appointment logbooks, and the absence of any notification system, are consistent with the pain points documented across these prior studies, further reinforcing the relevance and necessity of the proposed system.

2.2 Related Literature on Predictive Appointment Scheduling

The use of artificial intelligence and machine learning in managing healthcare appointments has been extensively studied. Salazar et al. conducted a systematic literature review of 24 articles on machine learning techniques applied to no-show appointments, finding that sending reminders to patients identified as high-risk by predictive models reduced no-show rates from 19.3% to 15.9% [1]. This directly supports the predictive scheduling feature proposed in this project, where the system is designed to flag at-risk appointment slots and trigger automated reminders to reduce wasted consultation time. At the IEEE level, a study published in IEEE Access proposed an integer linear optimization model that uses machine learning to estimate each patient's likelihood of attending their appointment, demonstrating that predictive

scheduling significantly improves resource utilization across multiple medical specialties [2]. Furthermore, Oikonomidi et al. conducted a rapid systematic review and found high-certainty evidence that predictive model-based text message reminders reduced no-shows with a median risk ratio of 0.91 meaning roughly 9% fewer missed appointments simply by sending AI-targeted reminders [3].

The literature in this area strongly supports the inclusion of a predictive scheduling module in this project, as the evidence consistently shows that Automated reminders and scheduling optimizations lead to measurable reductions in no-show rates across various healthcare settings. This is especially relevant given the client's direct acknowledgment during the interview that wasted appointment slots represent a significant operational and revenue problem for his clinic.

2.3 Related Literature on Electronic Health Records and Digitization

The shift from paper records to electronic health records (EHR) has been a major focus in healthcare informatics research globally. Closer to the context of this project, a Philippine-based systematic review by Derecho et al. examined EMR adoption in developing economies and identified usability, usefulness, and infrastructure as the key factors influencing adoption highlighting the need for simple and accessible systems that clinic staff with minimal technical background can operate [4]. Akwaowo et al. further found that perceived usefulness and anticipated success significantly drive clinician adoption of EMRs in developing countries, while data security concerns negatively affect it, underscoring the importance of building trust into the system design [5].

These findings directly informed the design priorities of this project. The clinic currently maintains thousands of physical patient folders that must be manually searched during each consultation, creating significant inefficiencies in record retrieval and contributing to the storage crisis the client identified. The emphasis on a user-friendly interface, secure data handling, and accessible web-based design in this project directly reflects the adoption factors identified in the Philippine and developing-country literature, ensuring that the system can realistically be operated by the clinic secretary without requiring specialized technical training.

2.4 Related Literature on Healthcare Automation and No-Show Prediction

Research from neighboring Southeast Asian countries provides particularly relevant context for this project. Hamdan et al. applied machine learning techniques to predict outpatient no-show appointments in a Malaysian tertiary hospital, confirming their effectiveness in an Asian healthcare setting closely comparable to the Philippine context of this study [6].

Taken together, these findings confirm that the demand for AI-driven clinic management tools is growing rapidly, and that the features proposed in this project predictive scheduling and digital

record management are aligned with current industry trends and evidence-based best practices. The absence of any automated scheduling or no-show management mechanism in the current clinic workflow represents a clear gap that this project is positioned to address.

2.5 Related Systems

Beyond academic literature, several existing systems provide useful points of comparison for this project. LocalMed is an online dental scheduling platform that integrates directly with practice management systems to eliminate double-bookings and scheduling conflicts [7]. While effective, LocalMed is designed for larger practices that already have existing digital infrastructure, making it inaccessible to small independent clinics like the one operated by the client in this project. Similarly, GoodCall offers a conversational AI assistant for dental appointment booking via phone, text, and web, demonstrating the feasibility of AI in dental scheduling [8]. However, it requires prior integration with existing software, which presents a barrier for clinics that are starting from zero. Finally, the October 2024 partnership between Open Dental and Bola AI to integrate voice-based periodontal charting into clinic workflows demonstrates the broader industry trend toward embedding AI directly into clinic management systems [9].

The gap identified across these existing systems is clear while AI-powered, AI-assisted, and digital clinic tools exist, none of them are designed specifically for small, independent dental clinics in the Philippines that lack existing digital infrastructure and technical staff. LocalMed and GoodCall both presuppose an existing practice management system as a prerequisite, while the Open Dental and Bola AI integration targets clinics already operating at a higher level of digital maturity. This project directly addresses that gap by designing a system from the ground up for a clinic that currently operates entirely on paper, ensuring that the solution is appropriately scoped, accessible, and deployable without any prior digital infrastructure.

Chapter 3: Current System

3.1 Technical Background

Record Keeping: Physical Paper Folders Patient charts and dental records are stored in physical folders organized manually by the secretary. There is no digital indexing or searchable database; retrieval is done by physically going through folders one by one.

Data Encoding: Microsoft Excel Patient information is manually transcribed from paper forms into an Excel spreadsheet on the clinic computer. The file is stored locally with no cloud sync or automatic backup, making it vulnerable to permanent loss in the event of hardware failure.

Appointment Management: Paper Logbook All appointment bookings, cancellations, and reschedules are recorded by hand in a physical logbook. There is no conflict detection or automated scheduling system, making double-bookings and scheduling gaps a recurring risk.

Patient Reminders: Manual Calls and Messages Appointment reminders are sent by the secretary through personal phone calls or text messages to each patient individually. The process is entirely manual, time-consuming, and inconsistent, with no automated trigger or tracking system in place.

Compliance: Physical Paper Records for BIR The clinic maintains all records in paper format to comply with Bureau of Internal Revenue requirements. No digital export or compliance tool exists; the secretary manually prepares and organizes paper documents for any audit or inspection.

Hardware: Single Clinic Computer The clinic operates on a single local computer used for Excel encoding and general administrative tasks. There are no networked devices, no server infrastructure, and no backup hardware available.

3.2 List of Processes

Paper-Based Records

1. Patient visits the clinic for a consultation.
2. The secretary manually retrieves the patient's physical folder from the storage area.
3. The secretary records the visit details by hand onto the paper chart.
4. The physical folder is returned to storage after the consultation.
5. Records are occasionally lost or misplaced with no recovery option.

Data Redundancy

1. The secretary receives new patient information or updated records.
2. The secretary manually writes the information onto a paper form.
3. The secretary then re-encodes the same information into an Excel spreadsheet on the computer.
4. Any updates require the same double-entry process to be repeated.

File Corruption Risk

1. The secretary encodes patient data into the computer.
2. Data is saved locally on the clinic computer with no backup system in place.
3. In the event of a computer crash or hardware failure, all locally stored data is lost.

4. There is no cloud backup or recovery process available to restore lost files.

No Automated Reminders

1. The secretary manually checks the appointment logbook for upcoming schedules.
2. The secretary personally calls or messages each patient one by one to remind them of their appointment.
3. If the secretary is busy or forgets, the reminder is not sent.
4. Patient misses the appointment due to no reminder, resulting in a wasted consultation slot.

BIR Compliance Constraint

1. Clinic attempts to shift toward digital record-keeping.
2. BIR audit or inspection requires submission of paper-based records.
3. Clinic must maintain physical paper copies of all records regardless of any digital system in place.
4. Fully paperless operation remains impossible under current BIR regulations.

3.3 SWOT Analysis

Strengths

- The clinic has an existing patient record system, even if paper-based, meaning staff are already familiar with the data being managed.
- Paper records require no internet connection, electricity beyond basic lighting, or technical knowledge to operate.
- BIR compliance is naturally met since paper records are already the format required by the Bureau of Internal Revenue.
- The secretary has developed a working routine around the current system, meaning day-to-day operations continue without technical disruptions.

Weaknesses

- Patient charts are stored manually and are vulnerable to physical loss, damage, or misplacement with no backup or recovery option.
- Records are hand-transcribed from paper to Excel, creating duplicate work that consumes significant secretary time each day.
- The current system has no protection against data loss from hardware failure, as the secretary herself expressed fear of losing data in a computer crash.
- There is no automated reminder system in place, requiring the secretary to personally call or message each patient, which is time-consuming and inconsistent.
- Manual appointment booking in a paper logbook creates a high risk of scheduling conflicts, double-bookings, and wasted consultation slots.

Opportunities

- Replacing the paper system with a digital solution would eliminate redundant paper-to-Excel transcription and free up significant secretary time daily.
- Automated SMS and email reminders can directly reduce no-show rates, which the clinic has already identified as a recurring operational and revenue problem.
- Cloud-based storage would eliminate the risk of data loss from computer crashes and provide a secure, retrievable backup of all patient records.
- A digital system would make patient records instantly searchable by name or ID, eliminating the need to manually go through thousands of physical folders.
- The growing push for healthcare digitalization in the Philippines creates a favorable environment for transitioning to a modern clinic management system.

Threats

- BIR regulations currently require paper records, meaning the clinic cannot fully abandon physical documentation regardless of how digital the new system becomes.
- Internet downtime at the clinic location could disrupt access to a web-based system during operating hours.
- The Data Privacy Act of the Philippines (RA 10173) imposes legal obligations on how patient data is stored, accessed, and protected, which must be addressed in any digital solution.
- Staff with unfamiliarity with new technology could slow adoption and temporarily reduce clinic efficiency during the transition period.

Chapter 4: Proposed Solution

4.1 Technical Background

Frontend: Electron Framework Based Desktop Application The system is built as a desktop application using the Electron framework, which allows web technologies to run as a native desktop application on the clinic's existing Windows computer. This approach eliminates the need for a separate browser and ensures the application behaves like a traditional installed program that the secretary and dentist are already comfortable using. Electron also allows the application to function with limited internet connectivity, displaying locally cached data when the connection is temporarily unavailable, and syncing with the cloud once the connection is restored.

Backend: REST API (Node.js) with MySQL or PostgreSQL, and Ruby API for PDF Generation The backend is powered by a REST API built on Node.js, which handles all data processing, business logic, and communication between the frontend application and the database. Patient records, appointment data, dental charts, and system logs are stored in a structured relational database using either MySQL or PostgreSQL, both of which are proven, widely supported database systems suitable for managing sensitive healthcare records. A separate Ruby API is integrated specifically for PDF generation, handling the creation of printable documents such as BIR-compliant forms, digital prescriptions, and patient record exports. Separating PDF generation into its own service ensures that document rendering does not affect the performance of the core application.

Cloud Storage: Google Cloud All clinic data is stored and backed up on Google Cloud infrastructure, which provides enterprise-grade security, high availability, and geographic redundancy. Automated daily backups are scheduled to run without any manual action from the secretary, ensuring that a full and recent copy of all patient records, appointments, and system data is always available for recovery. This directly eliminates the data loss risk the secretary identified, where a single computer crash could permanently destroy years of patient records stored only on the local machine.

Notifications: SMTP Email Reminders The system uses SMTP-based email to automatically send appointment reminders to patients at least 24 hours before their scheduled consultation. Reminder messages are generated and dispatched by the backend without any manual action from the secretary, and each message includes the patient's name, appointment date, appointment time, and the clinic's contact details. The system also flags any patient record with a missing email address and adds it to a follow-up list on the secretary's dashboard so that no appointment goes unnotified.

Hardware: Existing Clinic Computer the DCMS is designed to run on the clinic's existing computer hardware with minimal upgrades or additional purchases. The Electron-based desktop application is lightweight and optimized to perform reliably on standard office-grade hardware, ensuring that the clinic incurs zero additional capital expense on equipment as part of the system deployment. This was a deliberate design decision made in response to the clinic's operational context, where cost-efficiency and minimal disruption to existing infrastructure were identified as priorities.

Design Basis: Nielsen's 10 Usability Heuristics + ISO 9241-11 The user interface of the DCMS is designed in accordance with Nielsen's 10 Usability Heuristics and the ISO 9241-11 standard for ergonomics of human-system interaction. Nielsen's heuristics provide practical design principles such as visibility of system status, error prevention, and consistency, ensuring that the application behaves predictably and is easy to navigate even for users with minimal technical background. ISO 9241-11 defines usability in terms of effectiveness, efficiency, and satisfaction, providing a measurable framework against which the system's interface can be evaluated. Together, these standards ensure that both the clinic secretary and the dentist can operate the system confidently after a single orientation session, without requiring ongoing technical support.

4.2 Feasibility

Operational Feasibility

The system is operationally feasible as it directly addresses the Issues, confirmed pain points of the clinic's daily operations and is designed around the actual workflows of its users:
For the Secretary
The primary user of the system
Replaces manual paper logbooks with a digital appointment calendar
Eliminates the need to physically search through thousands of patient folders
Automated reminders replace manual patient follow-up calls
Welcomed the idea of automation during the client interview "She genuinely liked the idea of automating reminders"
For the Dentist
Can view the full day's appointment schedule in advance
Access complete patient records digitally during consultation
Predictive algorithm flags high no-show risk appointments early
For the Patients
Receive automated reminders before their appointments
No longer need to rely on personal memory for their schedule
Operational Fit
Designed for non-technical staff minimal training required
Limited to one clinic branch keeping the scope realistic and manageable
Does not interfere with BIR paper requirements as the system operates alongside existing compliance workflows

Economic Feasibility

Cost of the Problem (Current State)

Every cancelled or missed appointment results in lost consultation time and reduced clinic revenue confirmed directly by the dentist during the client interview ("Wala" — no system exists to recover wasted slots)

Secretary spends significant paid work hours manually calling patients, searching filing cabinets, and managing paper records

Physical storage for thousands of patient folders consumes clinic space and resources

Scheduling overlaps confirmed by the client result in disrupted operations and patient dissatisfaction

Benefits of the System (Proposed State)

Reduces unused consultation slots to no more than 2 per operating day directly recovering lost clinic revenue

Ensures at least 1 cancelled slot per day is reassigned to another patient maximizing consultation time

Eliminates the cost of physical storage all records stored digitally in a centralized database backed by Google Cloud

Frees up secretary time from manual tasks allowing staff to focus on higher value responsibilities

Reduces incomplete or missing patient records to no more than 1 occurrence per month reducing risks of poor patient care

Development and Operational Cost Considerations

Built as a desktop application requiring no special hardware beyond the clinic's existing workstation

No need for external system integration or additional infrastructure

Designed for a single clinic branch keeping development scope and cost manageable

Uses frameworks suitable for desktop development requiring no additional software licensing costs

Utilizes the Google Cloud Free Tier for initial academic deployment and testing bringing immediate hosting costs to PHP 0.00

Requires an estimated PHP 500 to PHP 1,000 per month for long-term Google Cloud database hosting and automated secure backups which is significantly cheaper than purchasing and maintaining a physical server

Commercial Viability and Cost Recovery

Features a highly scalable architecture allowing the team to commercialize and sell the desktop application to other small independent dental clinics for profit

Generates sustainable revenue through a software licensing model to fully cover ongoing Google Cloud hosting fees, database maintenance, and future updates post-deployment

Technical Feasibility

The system is technically feasible as it addresses clearly defined operational problems with proven and accessible technologies:

Desktop Application: Built using frameworks that support desktop development.

Centralized Database: Patient records, appointment history, dental and medical history are stored in a secure, centralized database eliminating the need for physical filing cabinets

Algorithm: Predictive Scheduling Machine learning analyzes past appointment patterns to predict no-show risk, a technique already validated in multiple healthcare studies including Malaysian and Philippine contexts

AI Assistant: An AI chatbot interface allows the dentist or secretary to query patient records, generate summaries and reports through natural language, compiled with the use of Llama

Automated Reminders: The system automatically sends appointment reminders to patients, replacing the secretary's manual follow-up calls

User-Friendly Interface: Designed for non-technical clinic staff, specifically the secretary, with minimal training required for daily operation

PDF Generation: A way to export Database Information in PDF to print, feasible with the use of Ruby API

Schedule Feasibility

The project is schedule-feasible given its clearly defined scope, limited to one clinic branch, and focused on two core modules. The estimated project timeline is 9 to 12 months:
Phase 1 — Requirements Gathering (Month 1 - Month 4)
Client interview
Follow-up interview with the clinic secretary
Pain points, workflows, and user stories documented
Phase 2 — System Design (Month 2 - Month 5)
As-Is and To-Be scenario maps
User stories for secretary, dentist, and patient defined
Empathy maps for all stakeholders completed
Phase 3 — Development (Month 6 - Month 9)
Desktop application development using selected frameworks
Centralized patient record database setup
AI Assistant integration for no-show prediction and chatbot
Phase 4 — Testing & Deployment (Month 8 - Month 9)
System testing with client
Deployment on clinic's existing workstation
Evaluate no show in clinic workflow along the system
Phase 5 — Monitoring & Evaluation (Month 8 - Month 9)
Monitor no-show rates and slot reassignment performance

Evaluate if objectives are being met
Apply corrections and improvements based on real clinic data

4.3 Requirements Analysis

4.3.1 Project Vision

Identity
The AI-Assisted Dental Clinic Management System is a desktop application developed specifically for small, independent dental clinics in the Philippines that currently operate entirely on manual, paper-based processes. The system was built in direct partnership with a licensed dentist who owns and operates a private dental clinic, ensuring that every feature and design decision is grounded in the real, confirmed operational challenges of the clinic and its staff.
The system is AI-assisted, meaning that artificial intelligence serves as a support layer that enhances the decision-making and workflows of clinic staff. The AI does not make decisions on behalf of users; it analyzes data, surfaces recommendations, sends automated notifications, and generates summaries to help the secretary and dentist act more efficiently. All final actions, including confirming bookings, reassigning slots, and approving records, remain under the control of authorized clinic staff.

Core Purpose
The system exists to solve four confirmed operational problems identified through client interviews and research:
The clinic loses consultation time and revenue every time a patient cancel or fails to show up, with no system in place to recover or reassign the vacant slot
The clinic has no mechanism to notify waiting patients when an appointment slot becomes available, preventing optimal use of available consultation time
Thousands of physical patient folders stored in filing cabinets makes the record retrieval slow, unreliable, and unsustainable as patient volume grows
The absence of a centralized digital record system makes it difficult and time-consuming for the dentist and secretary to quickly retrieve complete and accurate patient information during consultations and generate reports when needed

To address these problems the system delivers two core modules:
Patient Record Management Module A centralized, secure digital database that stores and organizes all patient information including personal details, dental history, medical history, past treatments, and appointment records. Records can be retrieved within five minutes during a patient visit, replacing the manual cabinet search process entirely. An AI chatbox is built into this module, allowing the dentist or secretary to ask the AI to summarize a patient’s information or

generate a full patient report directly from the stored records using natural language no technical knowledge required.
Predictive Appointment Scheduling Module A scheduling system backed by a predictive algorithm that analyzes each patient's appointment history to flag those at high risk of not showing up. The system automatically sends reminders to at-risk patients ahead of their appointments and alerts the secretary to act on flagged slots proactively. Cancelled slots are flagged instantly and made available for reassignment to waiting patients the same day, ensuring that no consultation time goes to waste.

Long-Term Objective
In the long term, this system aims to establish a sustainable, scalable digital foundation for the clinic's administrative operations one that can realistically be operated by non-technical staff with minimal training, grows more accurate over time as the predictive algorithm accumulates more appointment data, and sets a precedent for digital adoption in small, independent dental practices across the Philippines where manual clinic management remains the norm. The system also contributes to SDG 3: Good Health and Well-Being by improving the efficiency of dental care delivery, strengthening patient engagement through automated communication, and ensuring reliable continuity of care for every patient across every visit.

4.3.2 Prototype

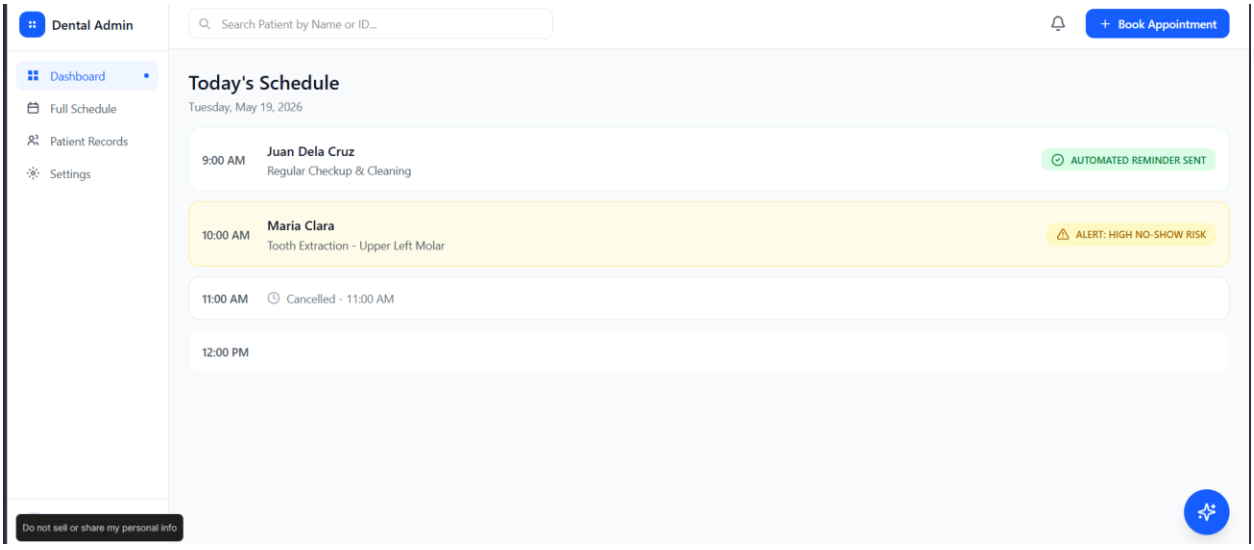


Figure 1: Figma Prototype Dashboard Page

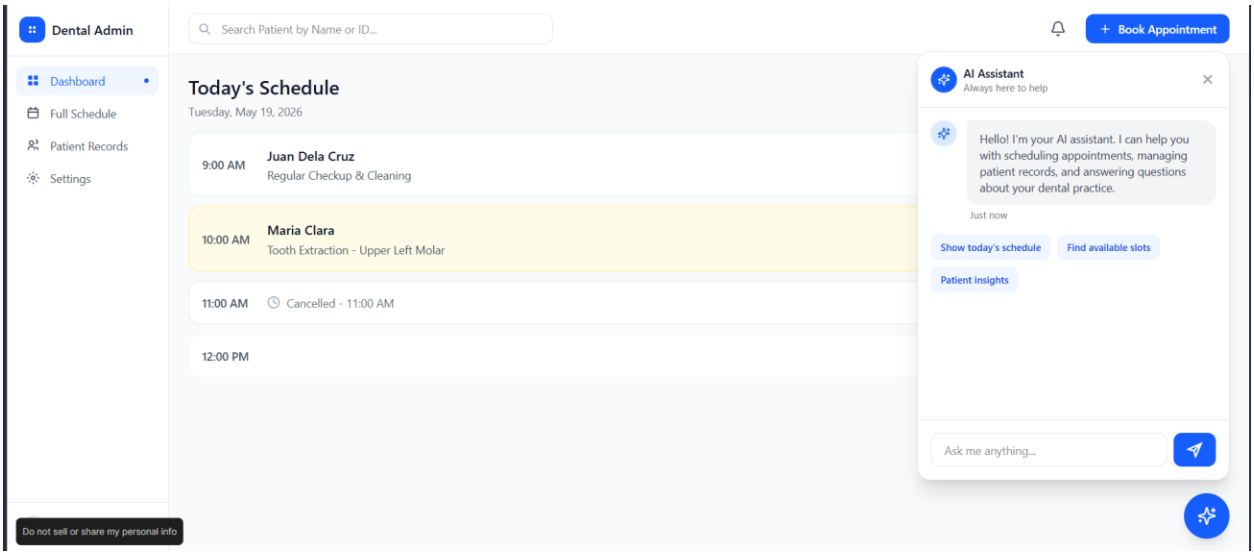


Figure 2: Figma Prototype AI Assistant Chatbot

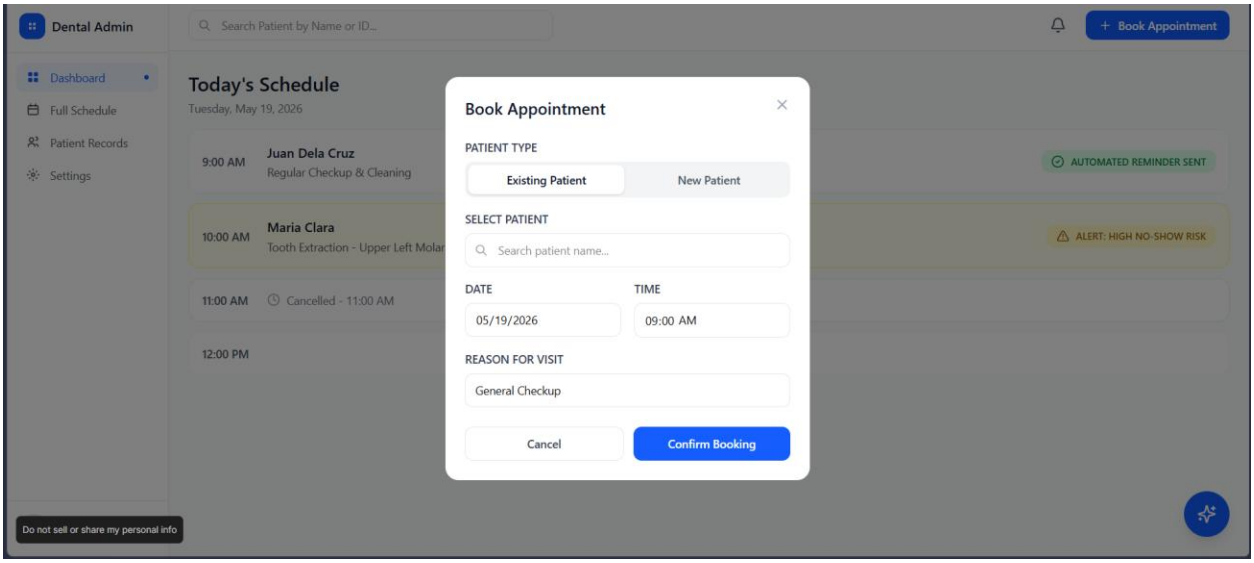


Figure 3: Figma Prototype Book Appointment Component

Appointment toggling for when the Patient has attended the clinic in the past, expanding the Patient Record. New Patients will be entered into the Patient Records.

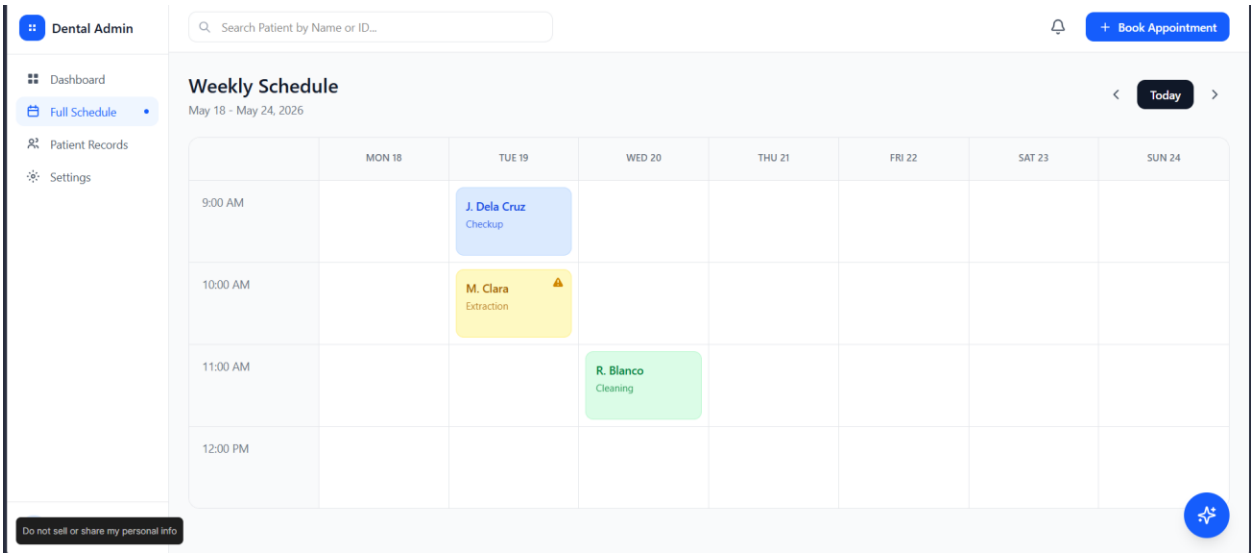


Figure 4: Figma Prototype Schedule Page

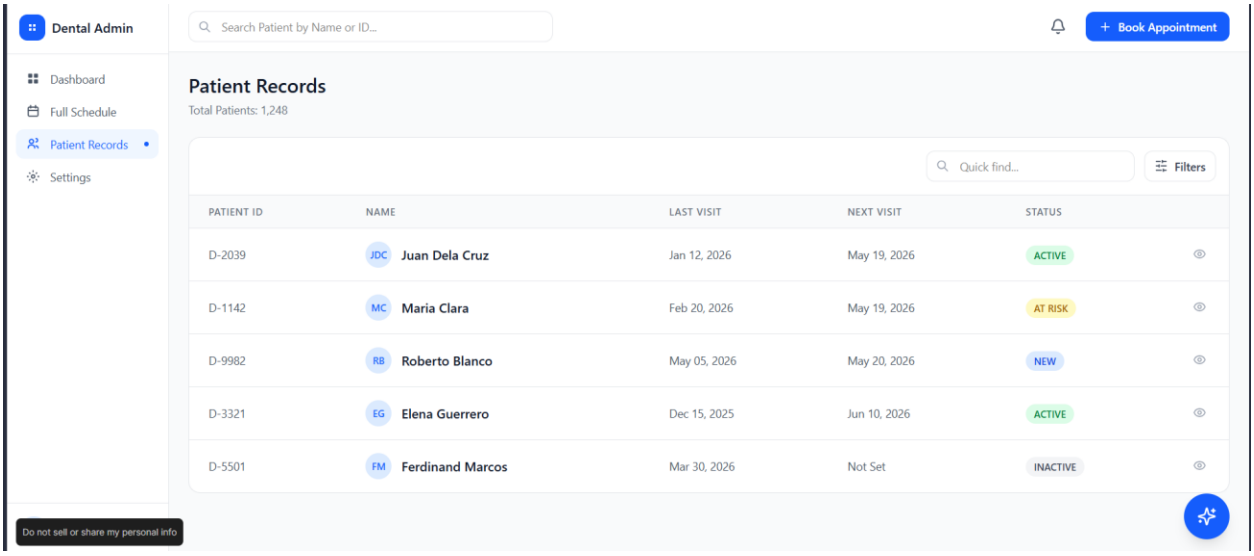


Figure 5: Figma Prototype Patient Record Page

The Patient Records can be viewed through the eye icon on the right.

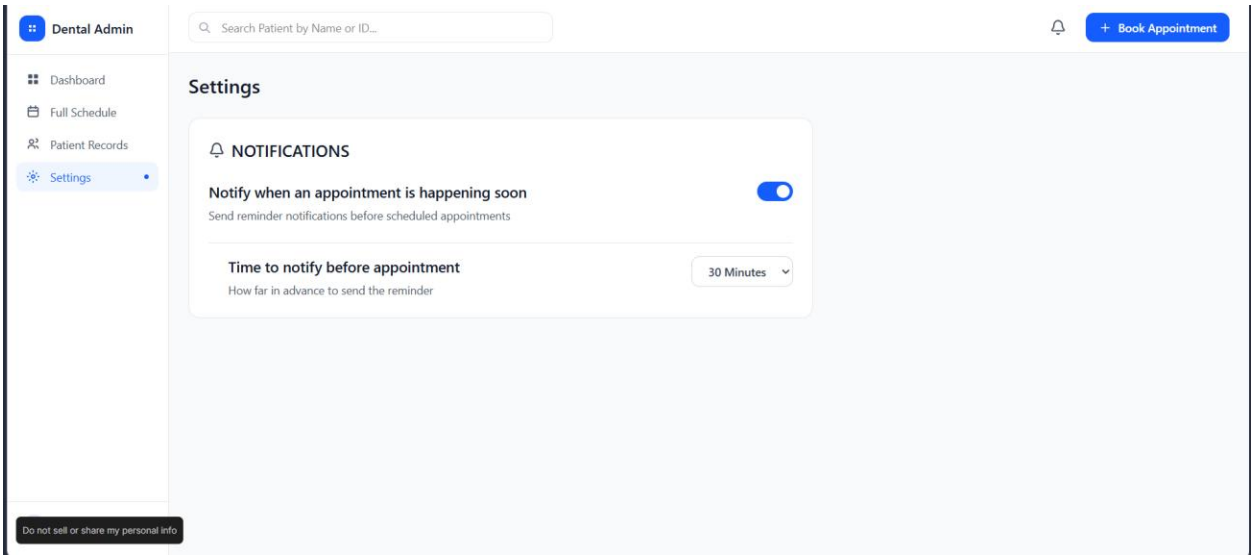
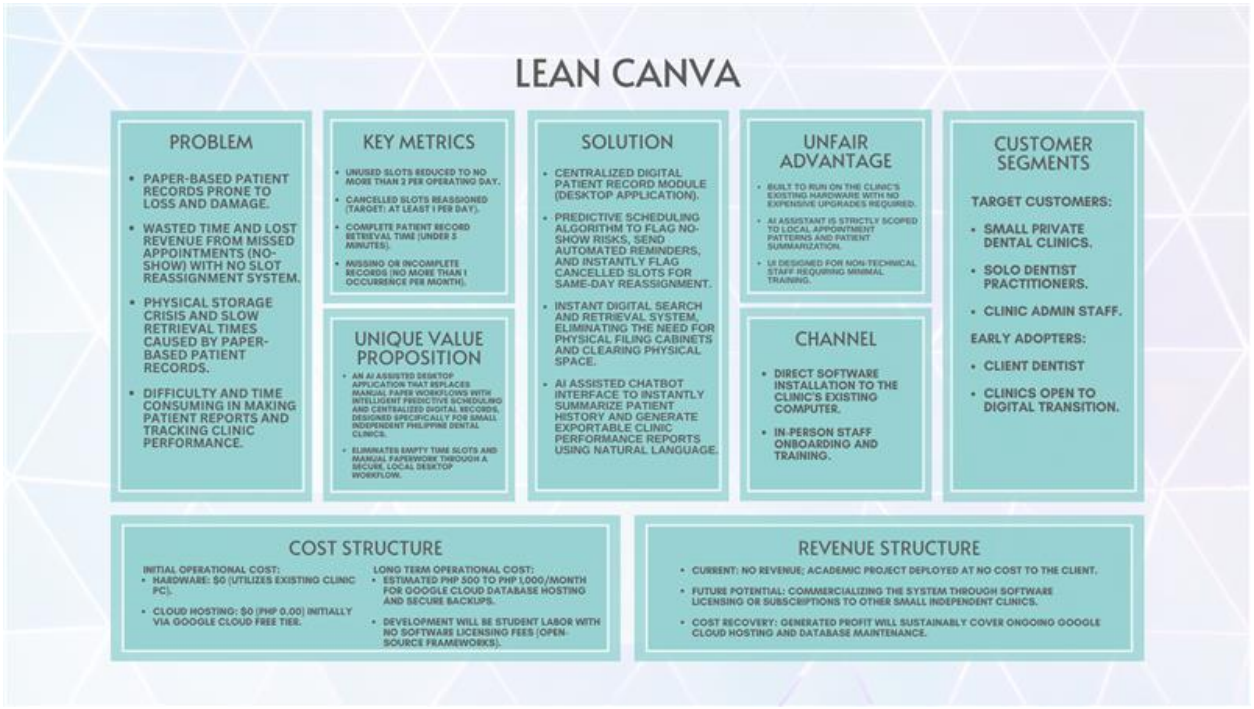


Figure 6: Figma Prototype Settings Page

This notification is separate from notifying the patient; this notifies the user if they are to expect someone soon.

4.3.3 Project Lean Canvas



4.3.4 User Classes and Characteristics

User Class	Attribute	Details
Secretary · Primary User		
Secretary	Description	The clinic secretary is the frontline administrator and the most frequent user of the system. She handles all daily operations appointment bookings, patient check-ins, record retrieval, and patient reminders. The system was designed primarily around her workflow to reduce manual, repetitive tasks confirmed during the client interview.
	Technical Proficiency	Low to Moderate familiar with Excel and basic computer use; no programming knowledge required
	Access Level	Full access to all administrative features; excludes clinical treatment note entry
	Primary Goals	Book and manage appointments digitally, retrieve patient records instantly, monitor dashboard alerts, reassign vacant slots, and receive automated reminder support
	Use Cases	UC-01, UC-02, UC-03, UC-06, UC-07, UC-09, UC-11
	Frequency of Use	Daily throughout all clinic operating hours
	Special Considerations	Interface must require minimal training. Expressed enthusiasm for automated reminders during interview ("She genuinely liked the idea of automating reminders"). Concerned about data loss from computer crashes.

Table 1

Dentist / Clinic Owner · Secondary User		
Dentist / Clinic Owner	Description	The dentist is the clinic owner who oversees all clinical operations. He accesses the system to review complete patient records before and during consultations, record treatment notes digitally, and monitor the daily schedule and no-show risk flags.
	Technical Proficiency	Moderate comfortable with digital tools; does not require a technical background to operate the system
	Access Level	Full system access including clinical treatment notes, AI chatbox, clinic reports, and schedule overview

	Primary Goals	View daily and weekly schedule, review complete patient history during consultations, record treatment notes digitally, monitor no-show risk flags, generate clinic performance reports
	Use Cases	UC-07, UC-08, UC-09, UC-10, UC-11
	Frequency of Use	Daily primarily during consultations and at the start and end of each clinic day
	Special Considerations	Confirmed wasted slots as a key pain point ("Wala"). Reacted positively to AI scheduling feature ("Maganda yan"). Manages two clinic branches; system covers one branch only.

Table 2

Patient · External User		
Patient	**Description**	Patients interact with the system externally. They receive automated booking confirmations and appointment reminders, and may access their own dental profile, appointment history, and treatment records. They have no access to administrative or clinical management features.
	Technical Proficiency	Varies system must be accessible to users with minimal digital experience across all age groups
	Access Level	Limited own profile, appointment history, and treatment records only; protected by personal login credentials
	Primary Goals	Receive appointment reminders, confirm or reschedule visits, and view personal dental and treatment records
	Use Cases	UC-01, UC-02, UC-05
	Frequency of Use	Occasional at time of booking, before scheduled appointments, and during post-visit follow-up
	Special Considerations	Patient data is protected under RA 10173 (Data Privacy Act of the Philippines). Login authentication is required for all personal record access.

Table 3

AI Assistant · System Actor

AI Assistant	**Description**	The AI Assistant is a system actor that supports clinic staff by performing automated background tasks within defined boundaries. It monitors appointments, calculates no-show risk, sends automated reminders, flags vacant slots for secretary review, and generates patient summaries and clinic reports via the AI chatbox (powered by Llama). It does not make decisions independently; all actions requiring judgment are surfaced to the secretary or dentist for confirmation.
	Technical Proficiency	N/A autonomous system actor; no human operation required for core background functions
	Access Level	Read and write access to appointment and patient data within defined system boundaries; cannot override confirmed bookings without secretary action
	Primary Goals	Predict no-show risk per patient, trigger automated reminders, flag and suggest slot reassignments, generate patient summaries and clinic performance reports on demand
	Use Cases	UC-03, UC-04, UC-05, UC-07, UC-08, UC-09
	Frequency of Use	Continuous runs in the background throughout all clinic operating hours
	Special Considerations	Prediction accuracy improves as more appointment history is collected. Operates on rule-based logic during early deployment until sufficient data exists for full AI model training.

Table 4

4.3.5 Product Backlog

Backlog Item ID	Type of Backlog Item	Description	Priority	Story Points	Status	Sprint	Acceptance Criteria
1	Feature	Digital appointment management system for the clinic	High	20	To Do	Sprint 1	The secretary can book, cancel, and reschedule appointments digitally with no double-booking.
2	User Story	As a secretary, I want to manage patient appointments digitally so	High	8	To Do	Sprint 1	Booking form saves correctly; occupied slots blocked; changes reflected

		that I can avoid scheduling conflicts and overlaps.					within 3 seconds.
3	User Story	As a secretary, I want to search for a patient record by name or ID so that I don't have to search through thousands of physical folders.	High	5	To Do	Sprint 1	Correct record returned within 5 seconds; full profile displayed without extra clicks.
4	User Story	As a secretary, I want to register a new patient into the system so that their information is stored digitally and retrievable for future visits.	High	3	To Do	Sprint 1	Patient saved with all required fields; unique ID generated; immediately searchable.
5	User Story	As a secretary, I want the system to automatically send reminders to patients so that I don't have to personally call or message each one.	High	5	To Do	Sprint 2	Reminder sent 24 hrs before appointment automatically; missing contact flagged on dashboard.
6	User Story	As a secretary, I want to see available slots after cancellations	High	5	To Do	Sprint 2	Slot marked available immediately; reassignable in under 3 clicks;

		and reassign them quickly so that no consultation time is wasted.					schedule updated in real time.
7	User Story	As a secretary, I want to cancel or reschedule an existing appointment so that the clinic schedule stays accurate at all times.	High	3	To Do	Sprint 2	Cancellation updates schedule instantly; freed slot available for rebooking; patient record updated.
8	User Story	As a secretary, I want to generate a summary report of appointments and no-shows so that the dentist and I can review clinic performance.	Mediu m	3	To Do	Sprint 3	Report generated for any date range; includes visits, appointments, and no-shows; exportable.
9	Feature	Patient records digitalization and dental chart management	High	20	To Do	Sprint 1	All patient records stored digitally; searchable; retrievable within 5 seconds.
10	User Story	As a dentist, I want to view a clear daily and weekly appointment schedule so that I can plan my day without surprises.	High	3	To Do	Sprint 1	Time-sorted schedule shows name, time, and type; auto-updates; daily/weekly toggle available.

11	User Story	As a dentist, I want to access a patient's complete dental and medical history digitally so that I don't rely on handwritten records during consultations.	High	5	To Do	Sprint 1	Full history shown chronologically in labeled sections; loads completely within 5 seconds.
12	User Story	As a dentist, I want to record treatment notes digitally after each consultation so that the patient's profile is always updated.	High	5	To Do	Sprint 2	Notes saved with timestamp and linked to patient ID; prior notes displayed in reverse order.
13	User Story	As a dentist, I want to know which patients are at high risk of not showing up so that I can prepare for possible empty slots early.	Medium	8	To Do	Sprint 3	High-risk flag shown on schedule for both dentist and secretary; 48-hr early reminder triggered.
14	User Story	As a dentist, I want to create and sign digital prescriptions so that patients receive a proper document instead of a photo of	Medium	5	To Do	Sprint 3	Prescription created and signed digitally; sent via email; printable format retained.

		handwritten paper.					
15	Non-Functional Requirement	System must load all records and schedule views within 5 seconds under normal operating conditions.	High	5	To Do	Sprint 2	All pages and record retrievals are completed within 5 seconds on the clinic's existing hardware.
16	Non-Functional Requirement	System must comply with the Philippine Data Privacy Act (RA 10173) for patient data protection.	High	5	To Do	Sprint 1	Patient data accessible only via authenticated login; access logs maintained; no unauthorized exposure.
17	User Story	As a patient, I want to book a dental appointment at a time that works for my schedule so that I can receive care without disrupting my routine.	Medium	5	To Do	Sprint 2	Available slots shown only; booking confirmed within 3 seconds; confirmation sent within 5 minutes.
18	User Story	As a patient, I want to receive an automated reminder before my appointment so that I don't forget and can reschedule early if needed.	High	3	To Do	Sprint 2	Reminder sent 24 hrs before; includes date, time, and clinic details; patient can confirm or reschedule.
19	User Story	As a patient, I want to reschedule my	Medium	3	To Do	Sprint 2	Reschedule saved after confirmation; original slot

		appointment when unforeseen circumstances arise so that I can still receive care at a later time.					freed immediately; confirmation sent within 5 minutes.
20	User Story	As a patient, I want to access my own dental profile and appointment history so that I can stay informed without calling the clinic.	Low	5	To Do	Sprint 3	Patient login grants access to own profile only; loads within 5 seconds; protected by authentication.

Table 5

4.3.6 Product Roadmap

Product Roadmap - AI-Enabled Dental Clinic Management System									
Feature / Deliverable	**Phase 1: Design**			**Phase 2: Development**			**Phase 3: Refinement**		
	Month 1	**Month 2**	**Month 3**	**Month 4**	**Month 5**	**Month 6**	**Month 7**	**Month 8**	**Month 9**
Requirements Gathering	Client Interview & Analysis								
Design Thinking & Empathy	Empathy Map	As-Is / To-Be Map							
Documentation	SOW & Lean Canvas	RRL & Paper Draft	Final Design Paper				Final Paper Update		Final Submission
UI/UX Design		Wireframes	UI Mockups & Prototype				UI Refinement		
System Architecture		Tech Stack Decision	System Architecture Design						
MILESTONE 1			Design Phase Complete - Paper & Prototype Approved						
Database Setup				Database Design & Setup					

Patient Record Module				**Build Record Module**	**Testing & Integration**				
Scheduling Module					**Build Scheduling Module**	**Slot Reassignment Feature**			
AI Chatbot					**Chatbot Basic Setup**	**Booking & Reminder Features**	**Report Generation**		
Notification System						**Auto Reminder & Alerts**			
MILESTONE 2						**Development Complete - Core Features Built & Integrated**			
System Testing							**Unit & Integration Testing**		
User Acceptance Testing							**UAT with Client & Secretary**	**Feedback & Bug Fixes**	
AI Model Refinement								**Train AI on Collected Data**	**Prediction Accuracy Check**
Deployment									**System Deployment & Handover**
MILESTONE 3									**Project Complete - System Deployed & Handed Over to Client**

Table 6

4.3.7 Release Plan

The release plan outlines the three planned development sprints for the AI-Assisted Dental Clinic Management System. Each sprint targets a specific set of product backlog items organized by priority and module dependency. The plan covers the core development phase (Months 6–9) as defined in the project roadmap.

Sprint	Timeline	Backlog IDs	Features / De liverables	Module	Story Points	Priority
Sprint 1 — Core Foundation (Months 6–7)						
Sprint 1	Months 6–7	PB-01	Digital appointment booking with conflict detection and real-time calendar	Appointment & Scheduling	8	High

		PB-02	Instant patient record search by name or ID; full profile retrieval	Patient Records	5	High
		PB-05	Daily appointment schedule view (daily / weekly toggle) for dentist	Appointment & Scheduling	3	High
		PB-06	Dentist access to complete digital dental and medical history during consultation	Patient Records	5	High
Sprint 1 Total Story Points: 21 — Release Goal: Core booking, record management, and schedule view operational						
Sprint 2 — Notifications, Cancellations & Treatment Notes (Month 7–8)						
Sprint 2	Months 7–8	PB-03	Automated 24-hour SMTP email appointment reminders with missing-contact flagging	Notifications	5	High
		PB-04	Cancelled slot flagging and same-day reassignment (3-click max)	Appointment & Scheduling	5	High
		PB-07	Digital treatment note recording with timestamp; reverse-chronological history display	Patient Records	5	High
		PB-09, PB-10, PB-11	Patient-facing booking, automated reminder receipt, and self-service reschedule	Appt. & Notifications	11	Med / High
Sprint 2 Total Story Points: 26 — Release Goal: Notification system, slot reassignment, and patient portal live						
Sprint 3 — AI Features, Predictive Scheduling & Patient Portal (Month 8–9)						
Sprint 3	Months 8–9	PB-08	No-show risk prediction model; high-risk flag on dashboard; 48-hour early reminder trigger	AI / Predictive Scheduling	8	Medium
		PB-12	Patient self-service portal: authenticated login, profile view, appointment history, and treatment records	Patient Records	5	Low
Sprint 3 Total Story Points: 13 — Release Goal: Full AI features integrated; system ready for UAT and deployment						

4.3.8 Use Case Diagram

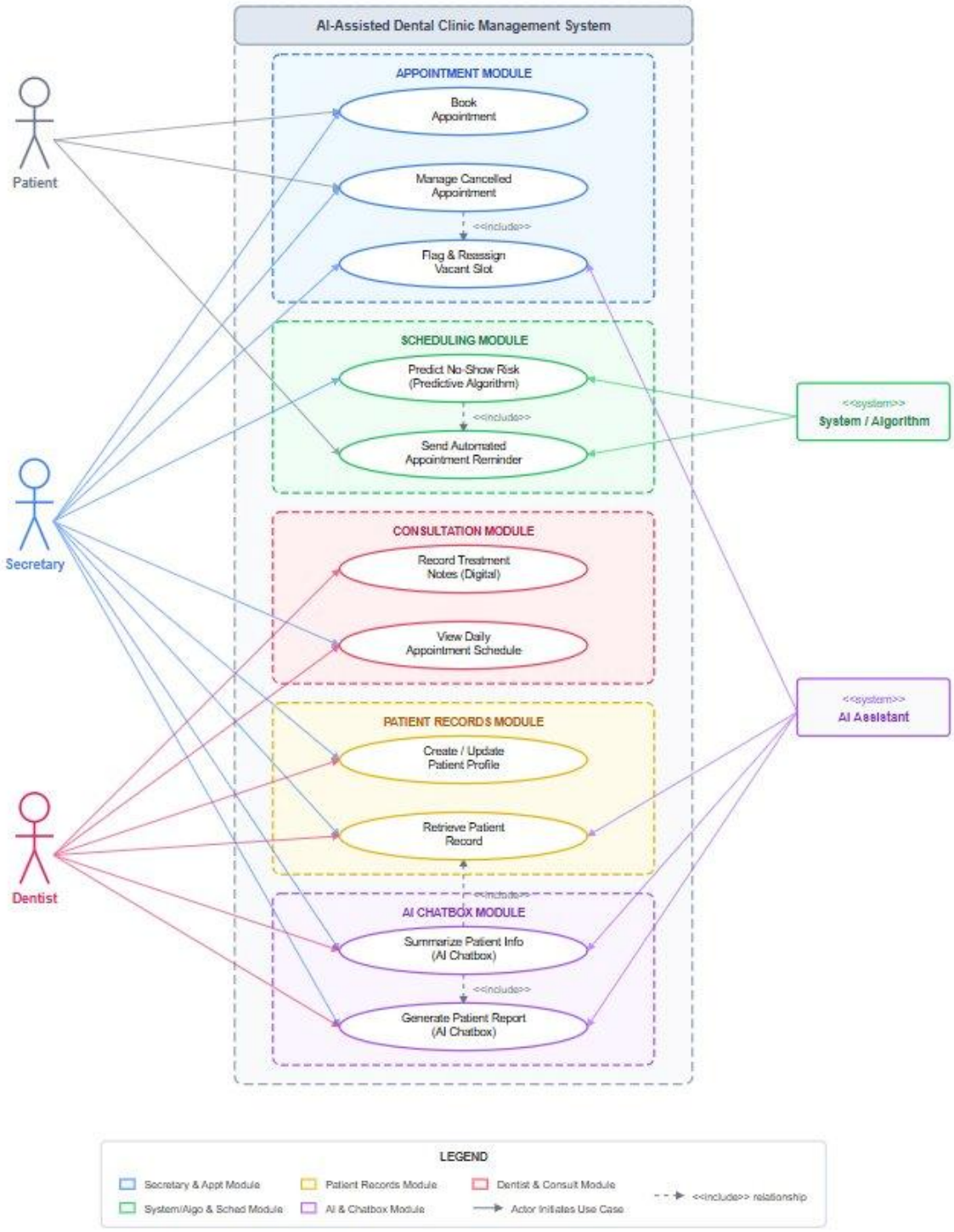


Figure 7: Draw.io Use Case Diagram

4.3.9 Use Case Full Description

UC-01: Book Appointment

Field	Details
Use Case ID	UC-01
Use Case Name	Book Appointment
Module	Appointment Module
Actors	Secretary, Patient
Description	The secretary books a dental appointment for a patient through the system

Pre-conditions:

- Secretary is logged into the system
- Patient contacts the clinic by phone or walk-in
- At least one appointment slot is available in the system

Post-conditions:

- Appointment is saved in the system
- Patient profile is created or updated
- Patient receives a booking confirmation

Happy Path (Basic Flow):

1. Patient contacts the clinic to request an appointment
2. Secretary opens the appointment module
3. Secretary searches for the patient in the system
4. If patient exists — secretary opens the existing profile
5. Secretary checks the real-time appointment calendar for available slots
6. Secretary selects the patient's preferred date and time
7. System confirms the slot is available
8. Secretary confirms the booking
9. System saves the appointment and links it to the patient profile
10. System sends an automatic booking confirmation to the patient

Alternative Flows:

Scenario	Steps
New Patient	At step 3, patients are not found in the system. Secretary creates a new patient profile by entering personal details, dental history, and medical history before proceeding to step 5
No Available Slot	At step 7, the preferred slot is unavailable. The secretary checks the next available slots and offers alternatives to the patient. Patient selects a new preferred time and flow continues from step 6

Patient Declines All Slots	Patient does not accept any available slots. Booking is not confirmed. Secretary notes the inquiry and ends the session

UC-02: Manage Cancelled Appointment

Field	Details
Use Case ID	UC-02
Use Case Name	Manage Cancelled Appointment
Module	Appointment Module
Actors	Secretary, Patient
Description	The secretary processes a patient's cancellation and updates the appointment status in the system

Pre-conditions:

- Patient has an existing confirmed appointment in the system
- Patient informs the clinic of the cancellation in advance
- Secretary is logged into the system

Post-conditions:

- Appointment status is updated to cancelled in the system
- Vacant slot is flagged as available
- Patient's profile reflects the cancellation
- Secretary is notified to take action on the open slot

Happy Path (Basic Flow):

1. Patient contacts the clinic to inform of cancellation
2. Secretary opens the appointment module
3. Secretary searches for the patient's upcoming appointment
4. System displays the confirmed appointment details
5. Secretary marks the appointment as cancelled
6. System updates the appointment status to cancelled
7. System automatically flags the slot as available
8. System notifies the secretary that the slot is now open for reassignment
9. Secretary proceeds to UC-03 to reassign the vacant slot

Alternative Flows:

Scenario	Steps

Patient Wants to Reschedule Instead	At step 5, patient requests reschedule instead of cancellation. Secretary searches for a new available slot and proceeds with UC-01 to rebook. Original slot is freed up
Appointment Not Found	At step 3, the appointment cannot be located. Secretary verifies patient identity and searches again. If still not found, secretary manually checks the logbook and updates the system
Last-Minute Cancellation	Patient cancels on the same day. Secretary marks as cancelled and system flags slot as urgent for same-day reassignment

UC-03: Flag and Reassign Vacant Slot

Field	Details
Use Case ID	UC-03
Use Case Name	Flag and Reassign Vacant Slot
Module	Appointment Module
Actors	Secretary, AI Assistant
Description	When an appointment slot becomes vacant due to cancellation or no-show, the system flags it and the secretary reassigns it to another patient

Pre-conditions:

- An appointment has been cancelled or a patient has not shown up
- The vacant slot has been flagged by the system
- Secretary is logged into the system

Post-conditions:

- Vacant slot is reassigned to another patient
- New patient receives a confirmation of the reassigned appointment
- Clinic productivity is maintained with no wasted consultation time

Happy Path (Basic Flow):

1. System detects a cancelled or vacant appointment slot
2. System automatically flags the slot as available on the secretary's dashboard
3. Secretary receives a notification of the open slot
4. Secretary reviews the waiting list for patients who can take the slot
5. Secretary contacts the waiting patient and confirms their availability
6. Secretary opens the appointment module and selects the vacant slot
7. Secretary assigns the slot to the waiting patient
8. System updates the appointment calendar with the new booking
9. System sends a confirmation notification to the newly assigned patient

Alternative Flows:

Scenario	Steps
No Waiting Patient Available	At step 4, no waiting patient is available. Slot remains open and is visible on the calendar. Secretary may attempt to contact other patients for the slot
AI Assistant Assists	The AI Assistant identifies the best matching waiting patient based on availability and urgency and suggests them to the secretary for confirmation
Patient Declines Reassigned Slot	At step 5, the waiting patient cannot take the slot. Secretary moves to the next patient on the waiting list and repeats steps 5 to 9

UC-04: Predict No-Show Risk

Field	Details
Use Case ID	UC-04
Use Case Name	Predict No-Show Risk
Module	Scheduling Module
Actors	System / Algorithm, Secretary
Description	The system algorithm analyzes historical appointment data to predict which patients are at high risk of missing their upcoming appointments

Pre-conditions:

- Sufficient historical appointment data exists in the system
- Upcoming appointments are scheduled in the system
- Scheduling module is active

Post-conditions:

- High-risk appointments are flagged on the secretary's dashboard
- Automated reminders are triggered for high-risk patients
- Secretary is alerted to monitor flagged appointments

Happy Path (Basic Flow):

1. System / Algorithm continuously monitors upcoming appointments
2. Algorithm analyzes each patient's appointment history — past no-shows, cancellations, and reschedules
3. Algorithm calculates a no-show risk score for each upcoming appointment
4. Appointments with a high-risk score are flagged on the dashboard
5. System automatically triggers UC-05 to send early reminders to flagged patients
6. Secretary is notified of high-risk appointments for manual follow-up if needed

7. If patient confirms — appointment status is updated to confirmed
8. If patient does not respond — secretary is alerted to reassign the slot via UC-03

Alternative Flows:

Scenario	Steps
Insufficient Data (New Patient)	At step 2, patient has no prior history. Algorithm assigns a neutral risk score. Standard reminder is sent at the default time
Early Deployment Phase	System has limited historical data. Predictive algorithm operates using basic rules until enough data is collected for full model training
Patient Confirms Immediately	After receiving the early reminder, the patient confirms right away. Secretary is notified and no further action is needed

UC-05: Send Automated Appointment Reminder

Field	Details
Use Case ID	UC-05
Use Case Name	Send Automated Appointment Reminder
Module	Scheduling Module
Actors	System / Algorithm, Patient
Description	The system automatically sends appointment reminders to patients ahead of their scheduled visits

Pre-conditions:

- Patient has a confirmed upcoming appointment
- Patient contact information is stored in the system
- Reminder schedule is configured in the system

Post-conditions:

- Patient receives a reminder notification
- Reminder status is logged in the system
- Secretary is alerted if no response is received

Happy Path (Basic Flow):

1. System detects an upcoming appointment within the reminder window
2. System / Algorithm determines the appropriate reminder timing based on the patient's risk level
3. System sends an automated reminder to the patient via email
4. Patient receives the reminder and is prompted to confirm or reschedule

5. Patient confirms their attendance
6. System updates the appointment status to confirmed
7. Secretary is notified that the patient has confirmed

Alternative Flows:

Scenario	Steps
Patient Does Not Respond	At step 5, patient does not respond within the set timeframe. System sends a follow-up reminder. If still no response, secretary is alerted to follow up manually
Patient Requests Reschedule	Patient replies requesting to reschedule. System collects preferred new time and triggers UC-01 and UC-02
High-Risk Patient	For high-risk patients flagged by UC-04, reminder is sent earlier than the standard window — 3 days before instead of 1 day

UC-06: Create / Update Patient Profile

Field	Details
Use Case ID	UC-06
Use Case Name	Create / Update Patient Profile
Module	Patient Records Module
Actors	Secretary, Dentist
Description	The secretary or dentist creates a new patient profile or updates an existing one in the centralized digital database

Pre-conditions:

- Secretary or dentist is logged into the system with authorized access
- Patient is present at the clinic, or their information has been provided

Post-conditions:

- Patient profile is created or updated in the system
- All changes are saved and timestamped automatically
- Updated profile is immediately accessible for consultation

Happy Path (Basic Flow):

1. Secretary opens the patient records module
2. Secretary searches for the patient by name or ID
3. If new patient — secretary selects "Create New Profile"
4. Secretary enters patient's personal details — name, contact, address
5. Secretary enters dental history and medical history

6. System saves the new profile and generates a unique patient ID
7. If existing patient — secretary opens the profile and selects "Update"
8. Secretary or dentist edits the relevant fields
9. System auto-saves all changes with a timestamp
10. Updated profile is immediately available for the current visit

Alternative Flows:

Scenario	Steps
Duplicate Patient Found	At step 2, system detects a possible duplicate profile. Secretary reviews both records and merges them if confirmed as the same patient
Incomplete Information	Patient cannot provide complete details at time of registration. Secretary saves partial profile and flags it for completion on next visit
Dentist Updates During Consultation	Dentist directly updates clinical notes and treatment findings in the patient profile during or after consultation

UC-07: Retrieve Patient Record

Field	Details
Use Case ID	UC-07
Use Case Name	Retrieve Patient Record
Module	Patient Records Module
Actors	Secretary, Dentist, AI Assistant
Description	The secretary or dentist retrieves a patient's complete digital record from the system for use during check-in or consultation

Pre-conditions:

- Patient has an existing profile in the system
- Secretary or dentist is logged in with authorized access

Post-conditions:

- Complete patient record is displayed on screen within 5 seconds
- Dentist can review full history before and during consultation
- No manual cabinet searching is required

Happy Path (Basic Flow):

1. Patient arrives at the clinic and states their name
2. Secretary opens the patient records module
3. Secretary types the patient's name or ID in the search bar
4. System retrieves and displays the complete patient profile instantly

Conclusion

This project presented the design and development of an AI-Assisted Dental Clinic Management System for a private dental clinic. The system was conceived in direct response to the operational challenges identified through a client interview and design thinking process specifically, the inefficient management of cancelled and missed appointments and the absence of a centralized digital system for patient records.

The study found that the clinic's current manual, paper-based operations are no longer sustainable given the growing volume of patients and records. The secretary's workflow is burdened by repetitive tasks such as manually searching through thousands of physical folders, personally reminding each patient of their appointments, and managing a paper logbook with no mechanism to handle cancellations or reassign vacant slots. As the client himself confirmed during the interview, when a patient does not show up, the slot simply goes to waste with no corrective action taken.

The proposed system directly addresses these problems through three core components: a digital patient record management module, an algorithm-driven appointment scheduling module, and an AI chatbot assistant. The digital record module eliminates the need for physical filing by centralizing all patient information in a secure, searchable database. The scheduling module uses a predictive algorithm that analyzes each patient's historical appointment patterns including past no-shows, cancellations, and reschedules to calculate a no-show risk score for each upcoming appointment. Based on this score, the system automatically sends reminders to at-risk patients and alerts the secretary to act on flagged slots proactively. Cancelled slots are flagged instantly and made available for reassignment to waiting patients the same day, ensuring that no consultation time goes to waste. The AI chatbot serves as the intelligent interface of the system, allowing the dentist or secretary to retrieve patient summaries, generate clinic performance reports, and interact with patient records through natural language no technical knowledge required.

The feasibility analysis confirmed that the system is operationally, economically, technically, and schedule-wise viable for the clinic's context. The interface is designed for ease of use by non-technical staff, the development timeline is distributed across three structured phases, and the predictive algorithm is designed to improve incrementally as more clinic appointment data becomes available.
It is acknowledged that the predictive algorithm's accuracy will be limited during early deployment due to the absence of sufficient historical digital data. However, this limitation is addressed by designing the system to operate on basic rule-based thresholds initially for example, flagging patients with two or more prior no-shows as high risk with the algorithm refining its scoring as appointment records accumulate over time.

Overall, this project demonstrates that even small, independent dental clinics in the Philippines can benefit significantly from digital transformation. By replacing paper-based processes with a centralized, AI-assisted platform, the clinic stands to recover lost consultation time, reduce the

administrative burden on the secretary, improve patient engagement through automated reminders, and gain data-driven insights into clinic performance all of which contribute to a higher quality of dental care delivery.

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Appendices

Appendix A: Roles and Responsibilities

Jon Joseph Catipon
Role: Leader
Responsibilities: Supervise the overall task of the project timeline and deliverables. Secure all outputs are on the track.

Prince Jabez Reyes
Role: Developer
Responsibilities: Ensures the system is working perfectly. Develop the interface of the system.

John Carlo Jancorda
Role: Technical/Developer Lead
Responsibilities: Leads the system architecture and development decisions. Reviews the code quality and the feasibility of all features

Joachim Emanuelle Tan
Role: Documentation Lead
Responsibilities: Responsible for the documentation of the project. Secures all documents are aligned with advisor and panelist feedback.

John Lourd De leon
Role: Tester
Responsibilities: Conducts a test for all system in the project. Gives feedback about the issues and documents of bugs.

Appendix B: Minutes of the Meeting

Attendees:

Joachim Tan

Prince Jabez Reyes

Jon Joseph Catipon

Joseph Jon Catipon (Client)

Date: 04/27/2026 Time: 8:00 pm

Discussion

The meeting was a client interview conducted with Joseph Jon Catipon, a dentist with two clinic branches, aimed at gathering information about the dental clinic's current business processes to support the development of the group's AI-assisted dental clinic management system.

Action

Joseph Jon Catipon explained the current patient intake process, which involves appointment scheduling through a secretary, arrival at an allotted time slot, filling up of patient information forms, and a follow-up interview inside the operating area.

Joachim Tan led the interview, asking questions about the clinic's scheduling process, handling of no-shows, and the possibility of implementing the product across the client's two clinic branches.

Prince Jabez Reyes presented the group's proposed product concept to the client, which includes an AI feature that predicts patient no-shows and sends automated scheduling reminders to patients based on past visit patterns.

Joseph Jon Catipon confirmed that the secretary currently handles all scheduling and patient follow-ups, and expressed that automation of this role would significantly reduce her workload.

The group requested permission to conduct a follow-up interview with the clinic secretary, to which Joseph Jon Catipon agreed, noting that it would need to be done in person at the clinic.

Minutes of the meeting

Adviser: Manuel L. Calimlim, Jr.

Attendees:

Joachim Tan

John Carlo Jancorda

Prince Jabez Reyes

Jon Joseph Catipon

John Lourd De leon

Manuel L. Calimlim, Jr.

8/4/2026

2:30 pm to 3:30 pm

Discussion

The meeting covered the revisions made to the paper and the daily updates and concerns related to the project.

Action

John Carlo Jancorda provided input regarding the context of one of the problem statements.

Jon Joseph Catipon helped clarify some of the identified problems with the client and consulted with Manuel L. Calimlim, Jr. on certain parts of the document.

Joachim Tan reported issues found in the paper and discussed them with Manuel L. Calimlim, Jr.

Prince Jabez Reyes reported issues found in the paper and discussed them with Manuel L. Calimlim, Jr.

Manuel L. Calimlim, Jr. raised concerns that the objectives in the document have not yet been updated and asked the group whether anything else is still missing. He also informed the group that he will be unavailable for advising in the coming days as he will be serving as a panelist. He advised the group to submit the paper by Friday for his review.

Minutes of the meeting

Attendees:

Joachim Tan

John Carlo Jancorda

Prince Jabez Reyes

Jon Joseph Catipon

John Lourd De leon

8/4/2026

9:00 pm

Discussion

The meeting covered the designated work to be done by each member of the group and revising the paper.

action

John Carlo Sends a link to post the finish work of each member of the group

Jon Confirms the schedule of the members who will interview the secretary and the content of the video teaser.

Joachim gives an tells the schedule for the teaser video and the some needed to be edited.

Prince gives and input idea for the content of the video teaser.

Minutes of the meeting (Interview to secretary) 23 min and 19 sec

00:00 Start

00:24 John Carlo questions secretary about handling the record of the patients

00:38 Secretary answered that a patient will go through her to do an appointment and finally scheduling

1:02 John Carlo asks another question about the hassles of using paper for making patient records and asks if there's a time where information is lost.

1:26 Secretart says "Minsan lang"/ sometimes

3:15 Prince asks about the storage of the records

3:20 Secretary said they currently use paper for storing patient records even though they used to store it digitally but reverted to paper.

3:39 Prince and John Carlo added on why they reverted to paper

3:48 Secretary said that the client (Clinic owner) decided to go back to paper

4:10 John Carlo asks if there's other papers the secretary fills up besides the patient records.

4:18 Secretary answers that there's no other forms besides the patient records.

5:33 Secretary stated that after finalizing the patient's records, she can put it in an excel spreadsheet instead of directly encoding it to the spreadsheet.

11:40 Secretary states a concern about the files being corrupted or deleted when the computer crashes.

11:52 Prince answers that when the file is saved on the storage (sdd) it won't be affected by computer crashes.

13:34 Secretary states that the requirement for BIR is still on paper.

16:10 John Carlo asks about how the secretary handles sudden cancellations

16:14 Secretary says it's going to be a free slot not unless someone walks in for an appointment

17:10 Secretary liked the idea of automating reminders to patients about their schedules at the clinic.

Appendix C: Methodology

As-Is Scenario Map						
PHASE / ACTOR	**STEP 1 Appointment Booking**	**STEP 2 Pre-Appointment**	**STEP 3 Patient Arrival**	**STEP 4 Record Retrieval**	**STEP 5 Consultation & Treatment**	**STEP 6 Post-Visit & Follow-up**

Patient Actions	• Patient calls the clinic by phone or walks in personally • Verbally tells the secretary their preferred date and time • Waits while secretary manually checks the paper logbook for an open slot • Receives verbal confirmation of their appointment no written or digital confirmation given	• Patient has no reminder from the clinic • Relies entirely on personal memory to remember their appointment date and time • No way to check or confirm their booking	• Patient arrives at the clinic • States their name verbally to the secretary at the front desk • Waits while the secretary manually searches through filing cabinets for their folder	• Continues waiting while their record is being located • May be asked to verbally repeat their dental history or medical history if the record is hard to find or incomplete • No digital record to refer to	• Patient proceeds to the operating area • Dentist interviews the patient and reviews the paper record • Undergoes dental examination and treatment • Receives verbal explanation of findings and treatment plan	• Patient leaves the clinic after treatment • No automated follow-up or next visit reminder is sent • Patient must personally call again to rebook their next appointment • No access to their own records or treatment history
Secretary Actions	• Answers the phone or attends to walk-in patient • Manually flips through the paper logbook to check for available slots • Writes the appointment details by hand in the logbook • No digital record of the booking is created	• No reminder is sent to the patient before their appointment • Has no way to know which patients might not show up • The reserved slot remains unconfirmed with no follow-up mechanism • Must personally call patients to remind them time consuming and inconsistent	• Greets the arriving patient • Manually searches through thousands of physical folders in filing cabinets one by one to locate the correct patient record • Process is slow especially during busy hours	• Retrieves the paper folder and hands it to the dentist or clinical staff • Updates the logbook manually if any changes are needed • If a record is misfiled or missing, the secretary must search again or ask the patient to re-fill information	• Waits outside the treatment area during consultation • Prepares any billing or payment records manually after treatment if needed	• Manually re-files the patient folder back into the cabinet after the visit • No automated system to schedule a follow-up or next appointment • No data is collected for tracking appointment patterns or patient retention

Dentist Actions	• Not involved in the booking process • Unaware of the day's full appointment schedule until the day itself	• Has no advance data on which patients might cancel or not show up • Cannot plan the day's workload ahead of time • Scheduling overlaps occasionally occur even under the current system confirmed by client during interview	• Reviews the paper record handed by the secretary before seeing the patient • May need to ask the patient to verbally repeat their history if the record is unclear, incomplete, or missing	• Must wait if the record is misfiled or cannot be found quickly • Relies on the patient's verbal recall when written records are unavailable • Risk of proceeding with incomplete patient information	• Conducts the patient interview in the operating area • Examines and treats the patient while referring to the handwritten paper record • Writes new treatment findings and clinical notes by hand directly onto the paper form in real time while the patient is in the dental chair • Verbally recommends next visit before patient leaves	• No digital log of treatment history is maintained • Cannot easily review patient trends, patterns, or past treatments across visits • Treatment continuity depends entirely on the paper folder being intact and correctly filed
Pain Points	• No digital confirmation of booking sent to patient • Double bookings or overlaps can occur confirmed by client • Logbook can be lost, damaged, or illegible	• High no-show rate due to no automated reminders • Secretary must manually call each patient inefficient and inconsistent • Wasted slots with no proactive system to prevent or reassign them • When patient cancels or no-shows' slot simply goes to waste (confirmed: 'Wala')	• Slow and tedious record retrieval searching cabinets one by one • Thousands of physical folders taking up massive floor space clinic is running out of storage • High risk of records being misfiled, lost, or damaged	• Missing or incomplete records slow down the entire consultation • Patient must repeat their history verbally inconsistent and unreliable • No centralized system to consolidate all patient data in one place	• Handwritten treatment notes are hard to read and inconsistent • No structured digital format for recording findings • Dentist cannot quickly reference past treatments during consultation	• No follow-up or next visit mechanism patient continuity is lost • No data available for clinic analytics, no-show tracking, or performance review • Secretary has no tool to monitor or improve patient retention

Opportunities (Proposed System)	• Digital booking with real-time slot availability • Automated booking confirmation sent to patient via SMS or notification • Eliminates double bookings and logbook errors	• Automated appointment reminders sent 24 hours and 1 hour before visit • No-show risk prediction flags high-risk appointments early • Secretary notified in advance to act on at-risk slots	• Instant digital record lookup by patient name or ID • No more searching through physical cabinets • All records compressed into a secure centralized digital database no physical space needed	• Complete and accurate patient profile always accessible • Dental history, medical history, and past treatments in one place • Secretary can pull up any record within 5 minutes during a visit	• Dentist records treatment notes digitally directly into the patient profile • Structured digital fields ensure consistent and readable documentation • Notes auto-saved after each session no paper charts needed	• Automated next-visit scheduling and reminders after each consultation • Cancelled slots flagged and reassigned to waiting patients the same day • Patient can access their own profile, appointment history, and treatment records anytime

Table 1

To-Be Scenario Map						
PHASE / ACTOR	**STEP 1 Appointment Booking**	**STEP 2 Pre-Appointment**	**STEP 3 Patient Arrival**	**STEP 4 Record Retrieval**	**STEP 5 Consultation & Treatment**	**STEP 6 Post-Visit & Follow-up**
Patient Actions	• Patient books appointment through the system via phone or web • Views available slots in real time no waiting for manual logbook check • Receives instant digital confirmation of their booking • Patient profile is created or updated automatically	• Patient receives an automated reminder notification on their phone before the appointment • Patient can confirm, reschedule, or cancel directly through the notification • No need to rely on personal memory system handles reminders automatically	• Patient arrives at the clinic • States name to secretary record pulled up instantly on screen • No waiting while cabinets are searched • Smooth and fast check-in process	• Patient record is already on screen when they arrive • No need to repeat dental or medical history verbally • All past treatments, history, and notes are visible and up to date • Patient proceeds to operating area immediately	• Patient undergoes dental examination and treatment • Dentist reviews complete digital patient profile during consultation • Patient receives clear explanation of findings based on documented history • Treatment is more informed and consistent across visits	• Patient receives automated next-visit reminder after the consultation • Can access their own profile, appointment history, and treatment records anytime • No need to call the clinic to rebook system prompts them when it's time

Secretary Actions	• Opens the system and checks real-time appointment calendar • Creates or updates patient profile digitally during booking • Books appointment with a few clicks no paper logbook needed • System automatically sends booking confirmation to patient	• System automatically sends reminders to all patients before their appointments • Secretary no longer needs to personally call each patient • System flags high-risk appointments so secretary can follow up if needed • More time freed up for other tasks	• Greets patient at the door • Searches patient name or ID in the system record loads in seconds • No more searching through physical filing cabinets • Check-in process is fast and organized	• Confirms patient details on screen and updates if needed • All records are centralized and secure no risk of losing or misfiling a folder • Secretary can handle multiple patients efficiently without delays	• Monitors the day's appointment schedule on the dashboard • Prepares next steps digitally after consultation • Any billing or payment notes can be added to the patient's profile	• System automatically suggests next appointment after treatment • Cancelled slots are flagged instantly secretary can reassign them the same day • All records auto-saved no manual re-filing needed • Secretary can track patient retention and appointment patterns through the dashboard
Dentist Actions (Joseph Jon Catipon)	• Not directly involved in booking • Can view upcoming appointments on the dashboard at any time • Knows the day's schedule in advance can plan workload properly	• System provides advance data on no-show risk for each patient • Dentist can prepare for possible empty slots ahead of time • No more scheduling surprises or overlaps	• Patient record is already pulled up and ready before the patient enters the operating area • Dentist can review complete history before the consultation begins	• Full patient profile available on screen dental history, medical history, past treatments, previous notes • No more relying on handwritten records that are hard to read • Dentist can make more informed decisions during treatment	• Dentist records treatment notes digitally during or after consultation • Notes are auto-saved to the patient's profile immediately • Structured digital format ensures consistent and readable documentation • Can review past treatments instantly during the session	• Complete digital treatment history maintained for every patient • Can easily review patient trends, patterns, and past treatments across visits • Treatment continuity is ensured regardless of how long ago the last visit was
System / AI Actions	• Real-time slot availability displayed automatically • Patient profile created or matched upon booking • Booking confirmation	• Automated reminders sent before appointment • AI analyzes patient history to predict no-show risk • High-risk appointments	• Patient record retrieved instantly upon name or ID search • System verifies patient identity and appointment	• Centralized database serves complete patient profile instantly • All records encrypted and securely stored • No physical	• Treatment notes auto-saved to patient profile after each session • System maintains complete and consistent	• Cancelled slots automatically flagged and made available for reassignment • System notifies waiting patients of

	sent to patient instantly	flagged and alerted to secretary in advance	details automatically	storage needed all data in the cloud	documentation automatically	newly available slots • Next visit automatically suggested based on treatment received • Data collected for clinic analytics and performance tracking
Improvements vs As-Is	✓ No more paper logbook ✓ Real-time availability ✓ Instant digital confirmation ✓ No double bookings	✓ No more manual reminders ✓ Automated notifications ✓ AI flags high-risk slots early ✓ Secretary time freed up	✓ Record retrieved in seconds ✓ No more cabinet searching ✓ Fast and smooth check-in ✓ No storage space needed	✓ Complete record always available ✓ No repeated history questions ✓ Secure centralized database ✓ No lost or damaged records	✓ Digital treatment notes ✓ Consistent documentation ✓ Full history accessible ✓ More informed consultations	✓ Cancelled slots reassigned same day ✓ Patient retention tracked ✓ Automated next-visit reminders ✓ Clinic analytics available

Table 2

User Stories

User Story 1: Secretary

As a clinic secretary, **I want** to manage patient appointments digitally, **so that** I no longer have to manually write bookings in a paper logbook and avoid scheduling conflicts and overlaps.

Acceptance Criteria:

1. The system provides a digital booking form where the secretary can input the patient name, preferred date, time slot, and appointment type, and saves the entry upon confirmation.
2. When a secretary attempts to book a time slot that is already occupied, the system highlights the conflict and prevents the booking from being saved until a different slot is selected.
3. Any booking, cancellation, or rescheduling action is reflected in the master schedule within 3 seconds without requiring a page reload.

As a clinic secretary, **I want** to search for a patient's record instantly by name or ID, **so that** I don't have to manually search through thousands of physical folders one by one just to find one file.

Acceptance Criteria:

1. The system provides a search bar that accepts both full name and partial name input, as well as patient ID, and returns all matching records ranked by closest match.
2. The full list of matching patient records is displayed on screen within 5 seconds of submitting the search query, regardless of the total number of records in the database.
3. The retrieved record displays the patient's complete profile including personal information, contact details, appointment history, and dental chart without requiring additional clicks to load each section.

As a clinic secretary, **I want** to have the system automatically send reminders to patients before their appointments, **so that** I don't have to personally call or message each patient myself to remind them.

Acceptance Criteria:

1. The system automatically generates and sends an SMS or email reminder to the patient exactly 24 hours before the scheduled appointment time without any action required from the secretary.
2. The reminder message includes the patient's full name, appointment date, appointment time, and clinic contact number.
3. If a patient record has no saved contact number or email address, the system automatically tags the appointment with a "No Contact Info" flag and displays it in a separate follow-up list visible to the secretary on the dashboard.

As a clinic secretary, **I want** to see when a slot becomes available after a cancellation and reassign it quickly, **so that** the empty slot doesn't go to waste, and another patient can take it the same day.

Acceptance Criteria:

1. When an appointment is cancelled, the system immediately updates the time slot status to "Available" and displays it in the open slots panel on the same scheduling screen without requiring navigation away.
2. The secretary can select the newly available slot and assign it to another patient in under 3 clicks from the moment the cancellation is confirmed.
3. Once reassigned, the slot displays the new patient's name, contact number, and appointment type in real time, and the previous patient's booking is moved to a cancelled records log.

User Story 2: Dentist

As a dentist, **I want** to see a clear overview of my daily appointment schedule, **so that** I know exactly how many patients are coming and can plan my day properly without surprises.

Acceptance Criteria:

1. The system displays a time-sorted list of all appointments for the current day, showing each entry's patient full name, scheduled time, and appointment type on a single screen.
2. When a new booking, cancellation, or reschedule is made by the secretary, the dentist's schedule view reflects the change automatically within 3 seconds without requiring a manual refresh.
3. The dentist can toggle between a daily view and a weekly view showing all appointments across seven days using a visible switch at the top of the schedule screen.

As a dentist, **I want** to access a patient's complete dental and medical history digitally during consultation, **so that** I don't have to rely on handwritten paper records that are hard to read or may be incomplete.

Acceptance Criteria:

1. Upon opening a patient profile, the system displays a chronological list of all past consultations including the date of visit, procedures performed, diagnoses recorded, and the name of the attending dentist.
2. All records are displayed in typed, legible text and are organized under clearly labeled sections such as Dental History, Medical History.
3. The complete patient profile including all historical records fully loads and is ready for interaction within 5 seconds of the dentist opening the profile.

As a dentist, **I want** to record my treatment notes digitally after each consultation, **so that** the patient's profile is always updated and I can easily review past treatments in future visits.

Acceptance Criteria:

1. After each consultation, the system provides an input form where the dentist can encode the diagnosis, procedure performed, materials used, and recommended next steps, and saves all entries upon submission.
2. Each saved entry is automatically timestamped, tagged to the correct patient ID, and stored in the patient's profile without overwriting or replacing any prior entries.
3. All previously recorded treatment notes for a patient are displayed in reverse chronological order within the patient's profile and are fully readable during any future consultation.

As a dentist, **I want** to know in advance which patients are at risk of not showing up, **so that** I can prepare for possible empty slots and the secretary can act on them early.

Acceptance Criteria:

1. The system automatically evaluates each booked appointment against the patient's history of missed or cancelled appointments and applies a "High No-Show Risk" flag to any appointment where the patient has missed two or more prior bookings.
2. Flagged appointments are displayed with a visible warning indicator in both the dentist's schedule view and the secretary's booking dashboard so both can act on them without switching screens.
3. When an appointment is flagged as high risk, the system automatically sends an additional reminder to the patient 48 hours before the appointment on top of the standard 24-hour reminder.

User Story 3: Patient

As a patient, **I want** to book a dental appointment at a time that works for my schedule, **so that** I can receive dental care without disrupting my daily routine.

Acceptance Criteria:

1. The system presents the patient with a calendar view showing only available time slots highlighted, and grays all fully booked dates and times to prevent selection of unavailable options.
2. Upon selecting a time slot and confirming the booking, the system saves the appointment and displays a confirmation screen showing the appointment date, time, and clinic address within 3 seconds.
3. The patient receives an SMS or email confirmation containing their appointment details, the clinic's contact number, and a reference number for the booking within 5 minutes of confirming.

As a patient, **I want** to receive an automated reminder before my appointment, **so that** I don't forget my schedule and inform the clinic early if I need to reschedule.

Acceptance Criteria:

1. The system sends an automated reminder message to the patient's registered contact number or email address exactly 24 hours before the scheduled appointment without any manual trigger.
2. The reminder message contains the patient's name, the exact date and time of the appointment, the name of the clinic, and the clinic's contact number for rescheduling.

3. The reminder message includes a reply option or link that allows the patient to confirm attendance or request a reschedule, and the secretary is notified of the patient's response on the dashboard.

As a patient, **I want** to reschedule my appointment when unforeseen circumstances come up, **so that** I can still receive my dental service at a later time without losing my slot entirely.

Acceptance Criteria:

1. The system allows the patient or secretary to initiate a reschedule request by selecting a new available slot from the calendar, and the change is saved only after explicit confirmation from the user.
2. Upon rescheduling, the original time slot is immediately released and marked as available in the system, making it bookable by another patient within seconds of the change being confirmed.
3. The patient receives an SMS or email notification containing the updated appointment date, time, and clinic details within 5 minutes of the reschedule being confirmed.

As a patient, **I want** to access my own dental profile and appointment history, **so that** I can stay informed about my treatments and upcoming visits without having to call the clinic.

Acceptance Criteria:

1. The system provides the patient with a personal login that grants access to a profile page displaying their full treatment history, list of past visits with dates and procedures, and all upcoming appointments.
2. The profile page presents information under clearly labeled sections and loads all content within 5 seconds of the patient logging in.
3. Patient profile data is protected by login authentication and is not accessible to any other patient account; only the patient themselves and authorized clinic staff with valid credentials can view or modify the records.

Empathy Map

Dental Dentist

SAYS

"Wala" when asked what happens when a patient doesn't show up

"Be on time. That's the only strict thing I have."

"Maganda yan" when we explained the AI scheduling feature

"Kahit ngayon nag-overlap pa rin ako" scheduling overlaps still happen

"We have to keep up with the times, of course"

"It'll be easier so patients can access their profile and send a ping to their cellphone"

THINKS

The current system is outdated and unsustainable

Wasted slots mean wasted money and time

A digital system would make the secretary's job much easier

Patients should be able to manage their own appointments

The clinic needs something simple enough for non-tech staff to use

Running two clinics makes organization even more important

DOES

Enforces a strict on-time policy for all patients

Conducts patient interviews in the operating area before treatment

Reviews handwritten paper records while patient is in the dental chair

Writes treatment notes by hand on physical dental charts

Manages two separate clinics with independent records and workflows

Relies on the secretary to handle all administrative tasks like appointments and scheduling

FEELS

Frustrated that cancelled slots just go to waste with no solution

Accepting of the current manual system but clearly aware it's not ideal

Excited about the possibility of AI managing schedules

Relieved at the idea of the secretary having a lighter workload and can prioritize more important tasks

Slightly overwhelmed managing two clinics without a digital system

Empathy Map (Dentist / Clinic Owner)

SAYS

"Wala" when asked what happens when a patient doesn't show up

"Be on time. That's the only strict thing I have."

"Maganda yan" when we explained the AI scheduling feature

"Kahit ngayon nag-overlap pa rin ako" scheduling overlaps still happen

"We have to keep up with the times, of course"

"It'll be easier so patients can access their profile and send a ping to their cellphone"

THINKS

The current system is outdated and unsustainable

Wasted slots mean wasted money and time

A digital system would make the secretary's job much easier

Patients should be able to manage their own appointments

The clinic needs something simple enough for non-tech staff to use

Running two clinics makes organization even more important

DOES

Enforces a strict on-time policy for all patients

Conducts patient interviews in the operating area before treatment

Reviews handwritten paper records while patient is in the dental chair

Writes treatment notes by hand on physical dental charts

Manages two separate clinics with independent records and workflows

Relies on the secretary to handle all administrative tasks like appointments and scheduling

FEELS

Frustrated that cancelled slots just go to waste with no solution

Accepting of the current manual system but clearly aware it's not ideal

Excited about the possibility of AI managing schedules

Relieved at the idea of the secretary having a lighter workload and can prioritize more important tasks

Slightly overwhelmed managing two clinics without a digital system

Secretary (Dental Clinic)

SAYS

"Minsan lang" — information loss from paper records happens, but only sometimes

Patients go through her for appointments and scheduling

They currently use paper for patient records, even though they previously stored records digitally before reverting

The clinic owner decided to go back to paper

There are no other forms to fill out besides patient records

After finalizing records, she can put data into an Excel spreadsheet rather than encoding directly

BIR requirements still mandate paper

Sudden cancellations just become a free slot unless a walk-in comes in

THINKS

Paper is the standard because that's what the clinic owner decided — it's not her call to change it

Digital storage is risky because computer crashes could corrupt or delete files

The Excel spreadsheet step feels like an acceptable middle ground between paper and full digital encoding

BIR compliance is a hard constraint that keeps paper in the workflow regardless of any digitization efforts

DOES

Manages patient appointments and scheduling as the front-of-the-process handler

Maintains patient records manually on paper

Transfers finalized paper records into an Excel spreadsheet

Handles schedule gaps caused by cancellations on the fly

FEELS

Somewhat resigned to paper-based processes since the decision came from above

Mildly concerned about data loss from paper (acknowledges it happens occasionally)

Noticeably worried about file corruption or deletion from computer crashes

Genuinely enthusiastic about the idea of automated patient reminders — it stood out as something she'd welcome

PAINS

Risk of information loss due to paper records, even if infrequent

No control over the reversion to paper — decision was imposed by clinic owner

Manual, redundant process of transcribing paper records into Excel

Fear of digital file loss from system crashes (though this was clarified to be less of a risk than she thought)

BIR compliance forces paper into the workflow regardless of any digital improvements

GAINS

Would benefit from automated reminders to reduce the manual effort of notifying patients

A more reliable digital records system could ease her concern about data loss

Clearer storage solutions (like cloud or SSD-based saving) could reduce crash-related anxiety

Story Board

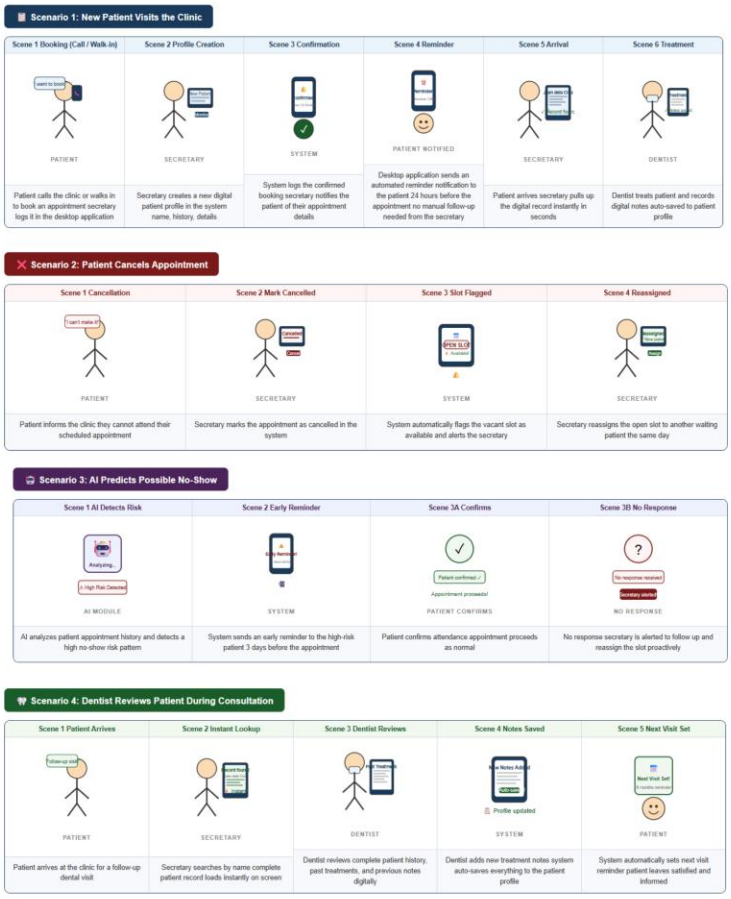
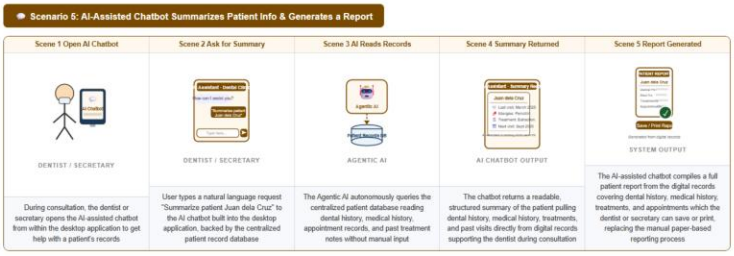

Scenario 1: New Patient Visits the Clinic
Scene 1 Booking (Call / Walk-in)
Scene 2 Profile Creation
Scene 3 Confirmation
Scene 4 Reminder
Scene 5 Arrival
Scene 6 Treatment
PATIENT
SECRETARY
SYSTEM
PATIENT NOTIFIED
SECRETARY
DENTIST
Patient calls the clinic or walks in to book an appointment secretary logs it in the desktop application
Secretary creates a new digital patient profile in the system name, history, details
System logs the confirmed booking secretary notifies the patient of their appointment details
Desktop application sends an automated reminder notification to the patient 24 hours before the appointment no manual follow-up needed from the secretary
Patient arrives secretary pulls up the digital record instantly in seconds
Dentist treats patient and records digital notes auto-saved to patient profile
Scenario 2: Patient Cancels Appointment
Scene 1 Cancellation
Scene 2 Mark Cancelled
Scene 3 Slot Flagged
Scene 4 Reassigned
PATIENT
SECRETARY
SYSTEM
SECRETARY
Patient informs the clinic they cannot attend their scheduled appointment
Secretary marks the appointment as cancelled in the system
System automatically flags the vacant slot as available and alerts the secretary
Secretary reassigns the open slot to another waiting patient the same day
Scenario 3: AI Predicts Possible No-Show
Scene 1 AI Detects Risk
Scene 2 Early Reminder
Scene 3A Confirms
Scene 3B No Response
AI MODULE
SYSTEM
PATIENT CONFIRMS
NO RESPONSE
AI analyzes patient appointment history and detects a high no-show risk pattern
System sends an early reminder to the high-risk patient 3 days before the appointment
Patient confirms attendance appointment proceeds as normal
No response secretary is alerted to follow up and reassign the slot proactively
Scenario 4: Dentist Reviews Patient During Consultation
Scene 1 Patient Arrives
Scene 2 Instant Lookup
Scene 3 Dentist Reviews
Scene 4 Notes Saved
Scene 5 Next Visit Set
PATIENT
SECRETARY
DENTIST
SYSTEM
PATIENT
Patient arrives at the clinic for a follow-up dental visit
Secretary searches by name complete patient record loads instantly on screen
Dentist reviews complete patient history, past treatments, and previous notes digitally
Dentist adds new treatment notes system auto-saves everything to the patient profile
System automatically sets next visit reminder patient leaves satisfied and informed

Figure 1



Figure

Appendix D: Project SharePoint Link

[Sharepoint Link]

Appendix E: Requirements Traceability Matrix (RTM)

Requirements Traceability Matrix (RTM)							
Req. ID	**Requirement Description**	**Source**	**Use Case(s)**	**User Story**	**Module**	**Priority**	**Status**
REQ-01	Secretary must be able to book a dental appointment for a patient through the system	User Story 1 (Secretary) Statement of the Problem	UC-01	US-SEC-01	Appointment Module	High	In Progress
REQ-02	System must prevent double bookings by detecting scheduling conflicts in real time	Client Interview SWOT Weaknesses	UC-01	US-SEC-01	Appointment Module	High	In Progress
REQ-03	System must send an automatic booking confirmation to the patient upon scheduling	User Story 3 (Patient) Objectives	UC-01	US-PAT-01	Appointment Module	High	In Progress
REQ-04	Secretary must be able to process a patient cancellation and update the appointment status	User Story 1 (Secretary) Statement of the Problem	UC-02	US-SEC-04	Appointment Module	High	In Progress
REQ-05	System must automatically flag a cancelled slot as available for reassignment	Client Interview ('Wala') Objectives	UC-02, UC-03	US-SEC-04	Appointment Module	High	In Progress
REQ-06	Secretary must be able to reassign a vacant slot to another patient within 3 clicks	User Story 1 (Secretary) Objectives	UC-03	US-SEC-04	Appointment Module	High	In Progress
REQ-07	System must notify the secretary when an appointment slot becomes open for reassignment	User Story 1 (Secretary)	UC-03	US-SEC-04	Appointment Module	Medium	In Progress
— AI SCHEDULING MODULE —							
REQ-08	System must analyze patient appointment history to calculate a no-show risk score per patient	Related Literature (Salazar et al.) Objectives	UC-04	US-DEN-04	AI Scheduling Module	High	In Progress
REQ-09	System must flag high-risk appointments on the secretary's and dentist's dashboard	User Story 2 (Dentist) Objectives	UC-04	US-DEN-04	AI Scheduling Module	High	In Progress
REQ-10	System must automatically send an appointment reminder 24 hours before a scheduled visit	User Story 1 (Secretary) User Story 3 (Patient)	UC-05	US-SEC-03 US-PAT-02	AI Scheduling Module	High	In Progress

REQ-11	System must send an additional reminder 48 hours early for high-risk flagged appointments	User Story 2 (Dentist) Related Literature	UC-04, UC-05	US-DEN-04	AI Scheduling Module	Medium	In Progress
REQ-12	Reminder message must include patient name, appointment date/time, and clinic contact number	User Story 3 (Patient)	UC-05	US-PAT-02	AI Scheduling Module	Medium	In Progress
REQ-13	System must log reminder status and alert secretary if no patient response is received	User Story 1 (Secretary)	UC-05	US-SEC-03	AI Scheduling Module	Medium	In Progress
REQ-14	For new patients with no history, system must assign a neutral risk score and send a standard reminder	Use Case UC-04 Alt Flow	UC-04, UC-05	US-DEN-04	AI Scheduling Module	Low	In Progress
—— PATIENT RECORDS MODULE ——							
REQ-15	Secretary or dentist must be able to create a new patient profile with personal, dental, and medical history	User Story 1 (Secretary) Statement of the Problem	UC-06	US-SEC-02	Patient Records Module	High	In Progress
REQ-16	System must generate a unique patient ID upon profile creation	Use Case UC-06	UC-06	US-SEC-02	Patient Records Module	High	In Progress
REQ-17	System must auto-save all profile changes with a timestamp	User Story 2 (Dentist)	UC-06	US-DEN-02	Patient Records Module	High	In Progress
REQ-18	Secretary or dentist must be able to retrieve a complete patient record within 5 minutes during a visit	Objectives User Story 1 (Secretary)	UC-07	US-SEC-02	Patient Records Module	High	In Progress
REQ-19	Search must accept full name, partial name, or patient ID and return results within 5 seconds	User Story 1 (Secretary)	UC-07	US-SEC-02	Patient Records Module	High	In Progress
REQ-20	System must detect and flag possible duplicate patient profiles during record creation	Use Case UC-06 Alt Flow	UC-06	US-SEC-02	Patient Records Module	Medium	In Progress
REQ-21	Incomplete patient profiles must be flaggable for completion on the next visit	Use Case UC-06 Alt Flow	UC-06	US-SEC-02	Patient Records Module	Low	In Progress
—— AI CHATBOX MODULE ——							
REQ-22	AI chatbot must generate a concise summary of a	User Story 2 (Dentist)	UC-08	US-DEN-02	AI Chatbox Module	High	In Progress

	patient's history and past treatments on request	Project Vision					
REQ-23	AI chatbot must answer follow-up questions about a patient's records using natural language	Use Case UC-08 Alt Flow Project Vision	UC-08	US-DEN-02	AI Chatbox Module	Medium	In Progress
REQ-24	AI chatbot must generate clinic performance reports on demand (e.g., no-show count, attendance rate)	User Story 2 (Dentist) Objectives	UC-09	US-DEN-04	AI Chatbox Module	High	In Progress
REQ-25	Generated reports must be exportable and downloadable as a PDF file	Use Case UC-09 Technical Background	UC-09	US-DEN-04	AI Chatbox Module	Medium	In Progress
REQ-26	AI chatbot must support custom report requests using natural language filters	Use Case UC-09 Alt Flow	UC-09	US-DEN-04	AI Chatbox Module	Low	In Progress
—— CONSULTATION MODULE ——							
REQ-27	Dentist must be able to record treatment notes digitally into the patient profile during or after consultation	User Story 2 (Dentist) Statement of the Problem	UC-10	US-DEN-03	Consultation Module	High	In Progress
REQ-28	Each treatment note entry must be auto-timestamped and linked to the correct patient ID	User Story 2 (Dentist)	UC-10	US-DEN-03	Consultation Module	High	In Progress
REQ-29	All past treatment notes must be displayed in reverse chronological order within the patient profile	User Story 2 (Dentist)	UC-10	US-DEN-03	Consultation Module	Medium	In Progress
REQ-30	System must display today's full appointment schedule on the dashboard upon login	User Story 2 (Dentist) User Story 1 (Secretary)	UC-11	US-DEN-01 US-SEC-01	Consultation Module	High	In Progress
REQ-31	Dashboard must highlight high-risk and flagged appointments visually	User Story 2 (Dentist)	UC-11	US-DEN-04	Consultation Module	High	In Progress
REQ-32	Dentist must be able to toggle between daily and weekly schedule views	User Story 2 (Dentist)	UC-11	US-DEN-01	Consultation Module	Medium	In Progress
—— SYSTEM / NON-FUNCTIONAL REQUIREMENTS ——							

REQ-33	System must be a desktop application installable on the clinic's existing workstation with no hardware upgrades	Scope & Limitations Technical Background	All UCs	All Stories	System-Wide	High	In Progress
REQ-34	System must require user authentication (login) for the secretary and authorized staff	Scope & Limitations	All UCs	All Stories	System-Wide	High	In Progress
REQ-35	All patient data must be stored securely in a centralized cloud-backed database (Google Cloud)	SWOT Threats Derecho et al.	UC-06, UC-07	All Stories	System-Wide	High	In Progress
REQ-36	System interface must be operable by non-technical clinic staff with minimal training	Operational Feasibility Derecho et al.	All UCs	All Stories	System-Wide	High	In Progress
REQ-37	System must comply with the Data Privacy Act of the Philippines (RA 10173) for patient data handling	SWOT Threats	All UCs	All Stories	System-Wide	High	In Progress
REQ-38	System must support PDF export of patient records for BIR paper compliance requirements	SWOT Threats Technical Background	UC-09	US-SEC-01	System-Wide	Medium	In Progress

Table

Appendix F: RACI Matrix

AI-Assisted Dental Clinic Management System						
R = Responsible | A = Accountable | C = Consulted | I = Informed						
Task / Deliverable	**Jon Joseph (Leader)**	**John Carlo (Tech Lead)**	**Prince (Developer)**	**Joachim (Docs Lead)**	**John Lourd (Tester)**	**Advisor Calimlim**
— PHASE 1: DESIGN —						
Client Interview & Requirements Gathering	A	C	C	R	I	C
Empathy Map & Design Thinking	A	C	C	R	I	C
As-Is / To-Be Scenario Mapping	A	C	I	R	I	C
User Stories Definition	A	R	C	R	C	C
System Architecture Design	A	R	C	I	I	C
Tech Stack Decision	C	R	R	I	I	C

UI/UX Wireframes & Prototype (Figma)	C	C	R	I	I	C
Lean Canvas & Project Vision	A	C	C	R	I	C
Product Backlog & Roadmap	A	R	C	R	I	C
Use Case Diagram & Descriptions	A	R	C	R	I	C
Paper Draft & Documentation	A	C	I	R	I	C
—— PHASE 2: DEVELOPMENT ——						
Database Design & Setup	I	R	C	I	C	C
Patient Record Module (Build)	I	A	R	I	C	I
Appointment Scheduling Module (Build)	I	A	R	I	C	I
Slot Reassignment Feature	I	A	R	I	C	I
AI Chatbot Basic Setup (Llama)	I	R	R	I	C	C
Automated Reminder & Notification System	I	A	R	I	C	I
No-Show Predictive Algorithm Integration	I	R	R	I	C	C
PDF Report Generation (Ruby API)	I	R	R	I	C	I
Frontend UI Implementation	I	C	R	I	C	I
—— PHASE 3: REFINEMENT ——						
Unit & Integration Testing	I	C	C	I	R	I
User Acceptance Testing (UAT) with Client	A	C	C	I	R	C
Feedback Collection & Bug Fixes	A	C	R	I	R	C
AI Model Training & Accuracy Check	I	R	R	I	C	C
Final Paper Update & Submission	A	C	I	R	I	C
System Deployment & Handover	A	R	R	I	C	C
Project Monitoring & Evaluation	A	C	C	R	R	C

Table

LEGEND	
R - Responsible	The person who does the work
A - Accountable	The person ultimately answerable for the task
C - Consulted	People who provide input and feedback
I - Informed	People kept up-to-date on progress

Table